



Practicing Model Pluralism

—
“From Expert Intuitions to Governance Infrastructure”

Abstract:

Veit's (2019) **strong model pluralism** shows that complex phenomena require multiple models serving different epistemic functions: explanation, prediction, and intervention are not optimized simultaneously. I argue these arguments extend to policy contexts, where AI governance faces analogous trade-offs that make pluralism permanent.

But how can governance proceed in practice given permanent pluralism? I develop three linked responses. *Diagnostically*: experts carry tacit “ensemble” knowledge that institutional formats compress into single-model outputs, creating pathologies (model-as-rhetoric and premature aggregation) that destroy the information responsible pluralism needs. *Methodologically*: uncertainty, information and models should be maintained in structured form until specific decisions force aggregation. *Conceptually*: “level-separation” shows that model-level pluralism is compatible with policy-level convergence — one can ask “which policies work well across models?” without conclusively determining “which model is best?”

These commitments derive specific requirements: ensemble maintenance, multi-functional queryability, and common representation. Pearl's (2009) causal Bayesian networks satisfy all three and AMTAIR (Automated Modeling of Transformative AI Risks; Meyer (2025a)) demonstrates feasibility by using LLMs to extract such argument networks from natural language sources. The contribution is not solving the “embarrassment of riches” (Grüne-Yanoff & Marchionni, 2018) but showing how pluralism can be *practiced* — made visible, navigable, and actionable for governance.

Author:

Valentin Jakob Meyer
Valentin.meyer@uni-bayreuth.de
Matr.: 1828610
Tel.: +49 (1573) 4512494
Philosophy & Economics Program
University of Bayreuth

Course:

Advanced Philosophy of Economics

Instructor:

Prof. Walter Veit

Word Count:

~ 6,500

December 31, 2025

Table of Contents

I. Introduction: Actionable Model Pluralism	1
I.1 Opening: Models Serve Different Functions	1
I.2 The Representation Answer	1
I.3 AMTAIR: Extracting Multi-Functional Representations	2
I.4 Thesis and Contribution	3
I.5 Roadmap	3
II. Theoretical Foundations: From Pluralism to Practice	3
II.1 Veit's Strong Model Pluralism	3
II.2 Why Pluralism Extends to Policy	5
II.3 The Underdetermination Foundation	5
II.4 The Selection Challenge and Veit's Response	5
II.5 Bayesian Model Averaging as Coordination Infrastructure	6
III. From Diagnosis to Practice	8
III.1 The Tacit Ensemble Phenomenon	8
III.2 Where Current Practices Fail	9
III.3 Level-Separation	9
III.4 The Uncertainty Preservation Principle	10
III.5 Capability Requirements for Pluralism	10
III.6 A Workable Representation	11
IV. AMTAIR: Demonstrating Feasibility	11
IV.1 System Architecture Overview	11
IV.2 Why Formalization Matters	12
IV.3 The Extraction Pipeline	12
IV.4 Policy Preference Innovation	13
IV.5 Three Emphases, Each Multi-Functional	14
IV.6 Validation and Quality	14
V. Critical Analysis: Limits and Challenges	14
V.1 Computational Complexity	15
V.2 Epistemic Culture Clashes	15
V.3 Automation Dependencies	15
V.4 Fundamental Boundaries	15
V.5 Technology-Enabled Practice	16
VI. Conclusion	16

Bibliography	18
---------------------	-----------

Appendices	i
Appendix A: ArgDown and BayesDown Specification	i
Appendix B: BMA Mathematics Extended	v
Appendix C: AMTAIR Technical Implementation	vii
Appendix D: Policy Preference Mapping	vii
Appendix E: Validation Methodology	viii
Appendix F: Abbreviations	ix
Appendix G: Glossary	ix
Appendix H: Abstract Argumentation Semantics	xi
Appendix I: Critical Analysis Extended	xiii
Appendix J: Technology-Enabled Philosophy	xv
Appendix K: Cross-Paper Dependencies	xvi
Acknowledgements	xvi

I. Introduction: Actionable Model Pluralism

I.1 Opening: Models Serve Different Functions

Walter Veit (2019, 2021) argues that scientific models serve a "multiplicity of functions" that cannot be simultaneously optimized. Some models explain *why* and *how* phenomena occur by revealing causal mechanisms. Others predict *what* will happen by maximizing empirical accuracy. Still others guide *which* interventions work by mapping actionable levers. Following Grüne-Yanoff & Marchionni (2018)'s terminology, we can distinguish explanatory (E), predictive (P), and interventionist (I) modes, and single models do not excel across all three simultaneously.

But if pluralism is permanent and different functions require different models, how can governance proceed? AI risks exhibit exactly the complexity that makes pluralism unavoidable: unprecedented technology, interacting social-technical systems, deep uncertainty about trajectories and interventions. Policy makers need to understand why AI systems might fail (E), forecast when dangerous capabilities emerge (P), and identify which regulations might actually work (I). Using multiple models is epistemically correct but practically challenging.

Consider a hypothetical policy debate on AI chip export controls. Commerce officials present one integrated economic model with a clear recommendation.¹ Across the table, AI safety analysts bring diverse models optimized for different purposes: alignment researchers with causal accounts of how optimization might go wrong², forecasters with scaling laws that predict capability trajectories³, and governance analysts mapping interventions and institutional levers⁴. The models do not converge because they are not aiming at the same goal.

The Commerce response is predictable: "You have dozens of models pointing in different directions. How can we make policy based on your confusion?" But this frames the problem incorrectly. The "confusion" stems not from model failure but from model *purpose*. Each model can succeed at what it is designed for; the challenge is that governance requires all three functions simultaneously.

The Commerce official's question deserves a better answer than "accept pluralism." That answer requires distinguishing two questions:

- **Model-level question:** "Which model is the best?" — Often unanswerable under genuine uncertainty
- **Policy-level question:** "Which policies perform well across models?" — Answerable given the models

This is the key insight: *model-level pluralism is compatible with policy-level convergence*. One can maintain diverse models, with different assumptions, emphases, causal structures, while still identifying interventions that are robust across them. The question is what makes this possible.

I.2 The Representation Answer

A standard response to Veit's pluralism is to accept it philosophically while treating it as a practical obstacle: selecting one model or simply averaging across them. But both risk collapsing pluralism prematurely — the first by discarding models, the second by hiding the structured disagreement that

¹Imagine a model projecting mid-term economic and geopolitical leverage from market dominance in advanced semiconductor supply chains, optimizing for export revenue, technology dependence ratios, and infrastructure control. Such a model might recommend eliminating safety export restrictions to maximize commercial profits.

²See for example Hubinger et al. (2019) on mesa-optimization and Christiano (2018) on iterated amplification challenges.

³Cf. Kaplan et al. (2020) and Branwen (2020).

⁴See Dafoe (2018) on governance challenges.

makes pluralism informative. I propose a different question: *What representation could support all three epistemic functions without collapsing the information prematurely?*

Causal Bayesian networks (CBNs) provide an answer. Pearl's (2009) framework provides multi-functional structure: **Explanation (E)** queries the causal graph structure ("Why does X happen?")⁵; **Prediction (P)** queries probability distributions ("How probable is Y ?"); **Intervention (I)** applies do-calculus ("What if we do Z ?"); interventions change the network structure, for example by severing incoming arrows. The same graphs serve all three epistemic goals. E/P/I are genuinely *functions* and *scientific goals* rather than model *types*.

This enables level-separation: with multiple models as CBNs, one can ask policy-level questions across models without first resolving which model is best.⁶

I.3 AMTAIR: Extracting Multi-Functional Representations

AMTAIR (Automated Modeling of Transformative AI Risks) demonstrates that candidate CBN formalizations can be extracted from natural language sources and arguments about AI risk (Meyer, 2025a, the author's master thesis). The system does not require experts to formalize their reasoning; it extracts the causal structure and models implicit in how they argue and communicate.

Recently made possible by frontier large language models, AMTAIR implements a two-stage pipeline:

1. **ArgDown extraction:** Arguments are parsed into hierarchical causal structures capturing claims and their relationships: what causes what, what supports what, what undermines what. This captures the *explanatory* structure of arguments and models.
2. **BayesDown enrichment:** Probability distributions are added: prior probabilities for base claims⁷, conditional probabilities for dependencies, uncertainty ranges for contested parameters. This specifies complete Bayesian networks supporting *prediction* and *intervention* analysis.

The resulting representations preserve multi-functional queryability. The same extracted network can be queried for causal explanations (Why might mesa-optimization emerge?), probabilistic predictions (What is the probability of catastrophic misalignment by 2040?), or intervention analysis (How would compute governance affect this probability?). The extraction process transforms natural language arguments into a common format without forcing selection or elimination of models.

Once models exist as explicit structures, they can be compared, analyzed, and queried, enabling the robustness analysis outlined in Sections II through IV. Making pluralism visible is not the end goal but the prerequisite for making it workable.

Automation proves essential. The predecessor project, MTAIR [Modeling of Transformative AI Risks; Clarke et al. (2022)], required manual formalization: 20+ expert hours per model, making comprehensive coverage of ensembles (sets of models) impractical.⁸ AMTAIR reduces extraction to minutes of computation plus 1-2 hours of validation per model. This cost reduction is not incidental; it is what makes proper uncertainty preservation feasible. When ensemble maintenance is cheap, you can aggregate late rather than early. This approach became practical only with frontier LLMs from 2023 and later (OpenAI, 2023).

⁵For a concrete example, think of the parent-child relationships between nodes such as in the Rain-Sprinkler-Grass example which will be discussed later.

⁶This does not dissolve Veit's diagnosis. Different models are still needed for different phenomena, and formalization risks losing nuance. But functional pluralism need not preclude structured policy analysis within a given domain.

⁷The "correct" or "rational" choices of prior probabilities are subject of heated debates in epistemology and decision theory which go beyond what can be covered in this paper.

⁸This may explain why similar conceptual frameworks have not gained wider research and policy uptake despite their theoretical and epistemic appeal. The effort required to manually formalize even a modest ensemble of a dozen models is in the hundreds of hours, a cost-benefit ratio that reduces adoption. Automation can fundamentally change this calculus.

I.4 Thesis and Contribution

The paper develops three linked contributions. The first is diagnostic: I argue that experts already carry implicit distributions over models built through years of practice (Section III.1), and that current institutional practices systematically destroy this information through premature model selection and aggregation (Section III.2). Single models in public discourse are communication artifacts, not reflections of genuine epistemic states; recognizing this reframes the Commerce official's objection as a symptom of institutional and procedural failures rather than a flaw in pluralism itself.

The second contribution is conceptual. I propose that epistemic uncertainty should be maintained in its richest available form, collapsed only when specific decisions require it (Section III.4). This uncertainty preservation principle, combined with level-separation,⁹ jointly imply specific technical requirements: ensemble maintenance, multi-functional queryability, and common representation (Section III.5). Pearl's CBNs, implemented as DAGs in BayesDown format¹⁰, satisfy all three and make level-separation operational rather than merely aspirational (Section III.6).¹¹

The third contribution is practical. AMTAIR demonstrates that such representations can be extracted from natural language sources at scale (Section IV). Extraction loses nuance, but it preserves the multi-functional queryability that matters. Feasibility is demonstrated; production-readiness is not claimed. This paper extends Veit's diagnosis into governance infrastructure: from why pluralism is unavoidable to how it can be made workable.

I.5 Roadmap

Section II develops theoretical foundations: Veit's strong model pluralism grounded in Levins' trade-offs and the multiplicity of functions, why these arguments extend to policy contexts, and how Bayesian Model Averaging enables coordination infrastructure and decision-making. Section III bridges diagnosis to practice: the tacit ensemble phenomenon, why current practice fails, the convergence of level-separation and uncertainty preservation, and the requirements this imposes, satisfied by Pearl's CBNs. Section IV demonstrates feasibility through AMTAIR. Section V critically examines what formalization risks losing, when level-separation fails, and what it means for this approach to emerge at a particular technological moment. Section VI concludes.

II. Theoretical Foundations: From Pluralism to Practice

II.1 Veit's Strong Model Pluralism

Veit (2019)'s strong model pluralism provides the theoretical foundation for understanding why complex phenomena, including AI risk, persistently require multiple models rather than convergence on a single framework or "theory of everything". His analysis rests on several interconnected arguments that challenge standard approaches to scientific modeling.

The Core Theses. Veit defends two interconnected claims. The analytical thesis holds that "any successful analysis of models must target sets of models, their multiplicity of functions within science, and their scientific context and history." Examining single models while ignoring their relationships with alternatives misses how models actually function in practice. The practical thesis extends this: "for almost

⁹"Induced monism" at the policy level, one might say, though the monism is provisional, query-specific, and preserves the underlying pluralism.

¹⁰The BayesDown format captures probability distributions over ArgDown graphs. It was developed by AMTAIR and will be introduced in Section IV.3; see Appendix A for full specification.

¹¹Philosophers may find versions of the proof based on Dutch-book theorems particularly engaging.

any aspect x of phenomenon y , scientists require multiple models to achieve scientific goal z ." This is not a temporary limitation awaiting better methods but a permanent truth.

Why Pluralism is Unavoidable. The permanence follows from several arguments. First, drawing on Levins (1966), models face irreconcilable trade-offs: "the contradictory desiderata of generality, realism, and precision" cannot be simultaneously maximized. Second, cognitive limitations meet world complexity: "a complex, heterogeneous nature and a mind that can only cope with few variables at a time" jointly ensure comprehensive models exceed human capacity. Third, models serve a "multiplicity of functions" (explanation, prediction, manipulation) and different functions require different optimizations. Fourth, model pluralism is inherently context-sensitive: what counts as a good model depends on specific purposes.

Formalizing the Multiplicity of Functions. Veit's "multiplicity of functions" can for example be formalized, following Grüne-Yanoff & Marchionni (2018)'s terminology, into three primary epistemic modes: explanation (E), prediction (P), and intervention (I).¹²

These goals cannot be simultaneously optimized: mechanistic detail (E) undermines predictive tractability (P); forecast accuracy (P) may require black-box methods sacrificing causal understanding (E); actionability (I) demands institutional focus abstracting from technical details.¹³

This generates what Veit calls permanent pluralism: the need for multiple models persists regardless of empirical progress. The field is not waiting for eventual convergence on the "true" model. Different purposes require different models, and complex phenomena like transformative AI require addressing multiple purposes simultaneously. The diversity is not a bug awaiting debugging through better science; it is an essential feature of dealing with multifaceted reality. Veit & Browning (2023)'s recent extension of this framework to neural networks confirms the distinction remains relevant for contemporary AI: deep learning models that predict well often explain poorly, while models capturing mechanisms well may lack quantitative accuracy.

Function vs Type

A relevant clarification: E , P , and I are examples of *epistemic functions* (modes of inquiry or scientific goals), not *model types* (essential categories) per se. The same representation can serve explanatory, predictive, and interventionist purposes depending on *how it is queried*.¹⁴

Pearl's (2009) CBNs provide exactly this multi-functional structure. The graph encodes causal relationships (E-queries), probability distributions enable inference (P-queries), and the do-calculus formalizes intervention reasoning (I-queries).¹⁵

The implication: rather than maintaining separate model libraries for E, P, and I purposes, governance can maintain CBNs queryable for all three purposes. Functional pluralism need not entail representational fragmentation.

¹²In AI safety: **Explanatory models (E)** maximize causal understanding — Hubinger et al. (2019)'s mesa-optimization work provides detailed causal accounts of how learned optimization might emerge, clarifying conceptual challenges but sacrificing quantitative precision. **Predictive models (P)** maximize empirical accuracy — Kaplan et al. (2020)'s scaling laws enable capability forecasting with remarkable precision, while Branwen (2020)'s synthesis articulates the broader scaling hypothesis. Yet these models often function as black boxes. **Interventionist models (I)** maximize actionability — governance frameworks for compute regulation identify verification mechanisms and implementation pathways, prioritizing feasibility over theoretical completeness.

¹³This echoes Cartwright (1999)'s arguments about the "dappled world": domains with different causal structures require different theoretical frameworks, and unification of science may be fundamentally misguided.

¹⁴Were E/P/I model types, one would need to merge a causal mechanism diagram with a black-box predictor with a policy flowchart, which seems impossible. Recognizing them as query modes transforms this into a tractable question: what representations support all three?

¹⁵Pearl's do-calculus distinguishes "what is $P(Y|\text{do}(X))$?" from mere conditioning because interventions sever incoming causal arrows — graph surgery that cannot be expressed in purely correlational representations. A regression model can tell you that compute and capabilities correlate; only a causal model can distinguish "restricting compute reduces capabilities" from "capabilities and compute share a common cause."

II.2 Why Pluralism Extends to Policy

Veit's arguments extend to policy contexts because they concern the *nature of complex phenomena*, as well as the sociology of science. Levins' trade-offs constrain *any* modeling of complex systems; cognitive limitations apply universally; and the multiplicity of functions is equally pronounced in policy, which simultaneously requires explanation, prediction, and intervention guidance.¹⁶

Veit himself gestures toward this extension: "the special roles of models within policy-making would support model diversity, even when the scientists within the field are committed to the idea that there is something like the 'best' model" (2019). If strong model pluralism is unavoidable for complex phenomena, and AI risk is a complex phenomenon, then AI governance faces irreducible, strong pluralism. The question is how to *work with* pluralism responsibly.

II.3 The Underdetermination Foundation

The philosophical necessity of pluralism gains additional support from the Duhem-Quine thesis (Duhem, 1954; Quine, 1951): hypotheses are never tested in isolation but only as conjunctions with auxiliary assumptions. When predictions fail, logic alone cannot deduce which component requires revision and what theories should be modified or discarded.

For AI governance, this underdetermination becomes particularly acute. Often researchers reason about systems that do not yet exist, using analogies from current systems that may not hold, projecting into regimes where interpolation becomes speculation. One cannot run controlled experiments on existential risks. Stanford (2006)'s analysis of "unconceived alternatives" deepens this concern: the history of science reveals that scientists systematically fail to conceive of hypotheses that later prove successful. Models not yet imagined may prove more important than disputes between existing frameworks.

This structural underdetermination and the high uncertainty make model pluralism epistemically mandatory. If evidence cannot uniquely determine correct models, maintaining multiple models is not a temporary concession but a required and permanent feature of the epistemic situation.¹⁷

II.4 The Selection Challenge and Veit's Response

After philosophical justification, strong pluralism faces a practical challenge. Grüne-Yanoff & Marchionni (2018) formalize what they call the "embarrassment of riches" problem: as model libraries expand continuously, selection becomes increasingly difficult. Their analysis of Rodrik (2015)'s diagnostic framework reveals structural challenges: procedures may not yield manageable candidate sets, or may require impractically many comparisons.

If one maintains dozens of AI risk models, how does one make decisions? The Commerce Department official's objection has force: multiple models without selection criteria appear to provide less guidance than a single focused model, regardless of pluralism's philosophical merits.

Veit's response reframes the entire problem. He argues that "model pluralism as a thesis about the relationship between science and nature undermines the very idea of a general model selection framework" (Veit, 2021). The embarrassment only exists if you seek one model for all purposes. But permanent pluralism means different models serve irreducibly different functions. The embarrassment stems not

¹⁶Policy contexts add further reasons: stakeholder diversity (different actors legitimately prioritize different aspects), accountability requirements (democratic legitimacy may favor explicable models), often irreversible choices and action under uncertainty. Most importantly, governance *requires* decisions and actions with real-world consequences at scale - and inaction is a choice too.

¹⁷Recent work in naturalized induction provides mathematical tools for navigating underdetermination. The Kantorovich-Rubinstein metric (Villani, 2009) offers a formal framework for measuring "distance" between probability distributions even when the true measure is only partially known, enabling comparison under radical uncertainty by asking "how different are these models' predictions?" rather than "which model is true?"

from having too many models but from expecting models to do what they are not designed for: serving all goals simultaneously. The infrastructure proposed here aims to preserve pluralism's epistemic benefits while recovering some of monism's practicability¹⁸: coordination without false consensus, action without premature selection.

Veit (2020)'s model anarchism raises a genuine challenge: if models are heterogeneous "all the way down," any common framework risks distortion¹⁹. The response lies in level-separation (Section III.3). This transformation requires formal mechanisms for model combination and policy evaluation that Veit's philosophy motivates but does not provide.

II.5 Bayesian Model Averaging as Coordination Infrastructure

Several frameworks exist for combining diverse models under uncertainty. Ensemble methods from machine learning (random forests, gradient boosting) have principled theoretical foundations and strong predictive performance, but their combination weights lack the interpretability that policy justification demands — a regulator cannot explain why a random forest assigned implicit weight 0.3 to one model.²⁰ Robust Decision Making [RDM; Lempert et al. (2006)] explores policy performance across scenarios without assigning probabilities — valuable when distributions are deeply contested, but limiting when evidence does support differential weighting. Delphi methods aggregate expert judgment through structured elicitation but do not capture the causal structure underlying disagreements. AMTAIR adopts Bayesian Model Averaging because it produces explicit, interpretable, revisable model weights grounded in probability theory — properties that governance contexts specifically require — though this choice may privilege quantifiable uncertainty over scenario-based or consensus-based approaches.

Bayesian Model Averaging (BMA) provides a mathematical framework for implementing pluralism formally (Hoeting et al., 1999).²¹ Rather than selecting a single model, BMA maintains the entire ensemble of models and weights predictions by model probability given observed data. This approach²² parallels ensemble methods now standard in climate science, where IPCC assessments routinely combine multiple climate models.

The core BMA equation is:

$$P(\Delta \mid D) = \sum_k P(\Delta \mid M_k, D) \cdot P(M_k \mid D)$$

Where Δ represents any quantity of interest, M_k represents individual models, and $P(M_k \mid D)$

¹⁸Even accepting pluralism's epistemic necessity, *acting as if* a single model were true conveys practical benefits: simplicity, tractability, clear communication, decisive action. The monist stance is epistemically incorrect but pragmatically tempting. Moreover, the aspiration toward unification can itself be scientifically productive: seeking a single comprehensive model drives theoretical refinement, exposes hidden assumptions, and reveals testable predictions, even when the aspiration remains fundamentally unrealizable. The goal is not to abandon this aspiration but to ensure that governance does not *depend* on or wait for its success.

¹⁹AMTAIR's response is not to deny this but to distinguish *perfect representation* (claiming to capture what models *are*) from *coordination artifacts* (enabling what models *do* in shared decision contexts). AMTAIR claims success on the latter: ArgDown and BayesDown provide minimal structure for comparison *without* claiming complete representational adequacy. The extracted networks capture the *predictive shadow* of each model, not its full content. That Veit's subsequent work (Veit & Browning, 2023) continues developing E/P/I for AI contexts suggests this bounded systematization is compatible with his broader framework.

²⁰This is not a deficiency of ensemble methods for their intended purpose (prediction), but a mismatch with governance contexts where decisions must be publicly justified and weights must be revisable in response to new evidence or democratic deliberation.

²¹For M-Open scenarios where no candidate model captures the true data-generating process, methods like Stacking (Yao et al., 2018) optimize combination weights for predictive performance rather than model truth, a potentially valuable alternative when the goal is decision support rather than truth discovery. AMTAIR's design can accommodate both approaches.

²²The approach aligns with philosophical arguments for model pluralism from thinkers like Cartwright (1999) and Mäki (2009) who emphasize that models function as tools for specific purposes rather than universal representations.

represents each model's posterior probability given data D . This weighted averaging²³ accomplishes several goals simultaneously: it preserves epistemic humility by refusing premature selection, enables action despite disagreement through weighted predictions, honestly quantifies uncertainty by incorporating model disagreement, and allows natural convergence as evidence accumulates. This local convergence on specific questions is compatible with permanent pluralism: even as individual uncertainties resolve for narrow questions, comprehensive coverage of complex phenomena like transformative AI always requires multiple models optimized for different epistemic goals.

Addressing the Monism Objection

A critic might object that BMA produces a single weighted output, sophisticated monism rather than genuine pluralism. The response distinguishes *prediction-level* outputs from *model-level* commitments.²⁴ All models persist with non-zero weight; none are eliminated; the ensemble evolves through Bayesian updating rather than selection or elimination. The output may be "singular" but the epistemic commitment is "plural": which model is correct remains unknown, all viable candidates are maintained, weights update as evidence accumulates.²⁵ This implements pluralism for coordination, not pluralism abandoned for tractability.

The Query Principle. BMA *enables querying across an ensemble* for specific answers while preserving model-level diversity. It is a tool for asking questions across models, not a permanent architecture for combining them. The principle: maintain pluralism and uncertainties as long as possible and condense the information only for the policy level when a specific question requires it.²⁶

The connection to Veit's diagnosis is structural. BMA's ensemble architecture directly addresses both the analytical thesis (models must be analyzed as sets, not individuals) and the practical thesis (multiple models are required for any complex phenomenon): all models persist with posterior weights $P(M_k | D)$, each contributing to weighted predictions without any being discarded. Permanent pluralism is respected because models are retained even with low weights; Bayesian updating adjusts influence without elimination. And because Levins' trade-offs mean no single model can maximize generality, realism, and precision simultaneously, BMA extracts value from differently optimized models without forcing any one to dominate. The multiplicity of functions is preserved not by BMA itself but facilitated by the information underlying the graph structures: each extracted network remains queryable for explanation (causal structure), prediction (probability distributions), and intervention (do-calculus and counterfactual modifications).

BMA's ensemble architecture presupposes model diversity: it is designed for situations where multiple models coexist and would be pointless with only one. This makes it a natural implementation framework for pluralism — the weighting mechanism respects models' different strengths on different questions, and

²³See Appendix B for extended mathematical treatment.

²⁴More precisely, $P(\Delta | D)$ is ideally a full (probability density) *distribution* over the quantity of interest, not just a point estimate. The "single output" characterization applies only when this distribution is collapsed to a summary statistic for decision-making. The full posterior preserves richer uncertainty information. Furthermore, prediction-level outputs need not be singular even for decision-making — decision theory accommodates distributions over outcomes, expected utility calculations, and sensitivity analysis *across* parameter ranges and scenarios.

²⁵A caveat: models can in principle be eliminated if *effectively Pareto-dominated* — contributing no unique information while adding computational cost. In practice, ensembles retain models with weights approaching zero rather than explicitly pruning but principled model retirement remains an open research question.

²⁶The workflow: (1) AMTAIR extracts individual models — full pluralism preserved; (2) models can be compared and analyzed — still no aggregation; (3) when a policy-level question demands an answer ("Is policy X robust in scenario Y ?"), BMA aggregates the information from all the different models *for that query*; (4) after the query, models remain distinct — no permanent collapse of diversity and information.

the aggregation formula combines²⁷ their contributions without forcing artificial convergence.²⁸

From Diagnosis to Practice. Veit's diagnosis is complete and strong pluralism is permanent. It is grounded in trade-offs that recur at every effort level, cognitive limits that cannot be transcended, and the multiplicity of functions complex phenomena demand. BMA provides infrastructure for working with multiple models, but this raises a prior question: how can governance proceed *practically* given permanent pluralism? What makes coordination possible without betraying the pluralist insights? Section III develops the bridge from diagnosis to practice.

III. From Diagnosis to Practice

Veit's diagnosis establishes that pluralism is permanent. But governance requires action. This section develops the bridge: how can governance proceed with permanent pluralism? The answer emerges from two converging insights: one about how expertise actually works, another about the level at which convergence should be sought, that together point to specific infrastructure requirements.

III.1 The Tacit Ensemble Phenomenon

How do experts actually practice pluralism? I suggest that experts carry something like implicit, intuitive distributions over models, built through years of practice, that they do not fully articulate. Consider an AI safety researcher asked the dreaded question, to estimate $P(\text{existential catastrophe from AI by 2100})$. Their response — "15-20%" — is not a reading from any single model but conceals a complex epistemic state weighting multiple scenarios.²⁹ For these kinds of questions researchers seem to implicitly compute something resembling their probability distribution over the outcomes given their (world-)models.³⁰ Whether experts actually perform implicit BMA or use fast-and-frugal heuristics that produce similar outputs is contested — Gigerenzer et al. (1999) argues experts may succeed *despite* not computing BMA, through ecologically adapted heuristics rather than probabilistic reasoning.³¹ The functional claim is what matters: expert judgment reflects multi-model consideration, and this structure is worth preserving regardless of the underlying cognitive mechanism.

When experts are challenged on a single model, it often turns out that the specific model had never been the actual basis of the judgment — it was a *placeholder* for deeper ensemble understanding based on their aggregate experience, intuitions, and judgments. Single models are communication artifacts, incomplete representations of what experts actually know. The question becomes: why is this ensemble knowledge so regularly lost in policy contexts?

²⁷Practical BMA implementation raises several technical challenges: handling model correlations when models share theoretical foundations or empirical data, assigning prior model probabilities (see Berger (2006)), statistical implementation methods (see Hansen et al. (2011)), and achieving coherence under computational constraints. These are covered in Appendix B.

²⁸Such strong model pluralism gains additional support from social epistemology. Zollman (2010)'s formal models demonstrate that scientific communities maintaining diverse approaches longer often achieve better outcomes than those converging quickly. Heesen et al. (2019) extend this to methodological triangulation: agreement across genuinely independent methods provides stronger evidence than any single approach. For AI governance, where crucial evidence may emerge only after irreversible decisions, BMA serves as infrastructure preserving alternative models against standardization pressures.

²⁹Maybe something like: Bostrom-style fast takeoff scenarios suggest 30-40%; Christiano-style slow takeoff scenarios suggest ~10%; optimist scenarios suggest 5-10%; each weighted by a web of beliefs and credences the researcher cannot fully articulate explicitly.

³⁰ $P(\text{outcome}) = \sum_i P(\text{outcome} \mid \text{Model}_i) \cdot P(\text{Model}_i \text{ is relevantly correct})$

³¹The claim here is functional, not mechanistic: whatever cognitive process experts use, their judgments reflect consideration of multiple models, and BMA provides a formal framework for representing this structure. The framework's value does not depend on BMA being the true cognitive process — it depends on capturing the *structure* of multi-model reasoning that institutional formats currently destroy.

III.2 Where Current Practices Fail

A tacit ensemble cannot be communicated when institutional formats force single-model outputs: congressional testimony, advisory reports, expert elicitation all take distributions over models and force the output "point estimates".³²

This worsens two pathologies. **Model-as-rhetoric**: experts select the single model that best supports their view and present it as *the* basis for their recommendation. Public debate becomes model-versus-model, a shadow play neither side fully believes. Policy makers see "contradictory expert recommendations" and overlook the actual constraints known by experts.

Premature aggregation: models (if multiple models are used at all) are aggregated and averaged long *before* being queried for specific questions.³³ If BMA produces a single standing prediction rather than being queried per-decision, model identity is lost, disagreement structure is lost, the ability to ask different questions differently is lost.³⁴

Consider congressional testimony: three experts present contradictory recommendations. A senator interrupts: "Are you saying the other two are *wrong*?" The truth: each expert's "position" is a single-model over-simplification; each also holds more complex views. But such formats force the contradiction.

III.3 Level-Separation

The diagnostic insight (III.1-III.2) reveals that current practice fails by forcing model-level selection or lossy aggregation. But this suggests a different approach: what if the required convergence were sought at a different level and step? Distinguish two levels: **Model-level question** ("Which model is the best?") — often unanswerable; **Policy-level question** ("Which policies perform well across models?") — answerable given the models.

This is *level-separation*: model-level pluralism is compatible with policy-level analysis. One can maintain diverse models while asking "which policies are robust across models?" without first resolving "which model is the best?" The Commerce official's question assumed deciding requires model-level convergence first. Level-separation shows this assumption is false.

Level-separation is *not* selecting the best model, averaging models into false consensus, ignoring disagreement, or relativism (some policies genuinely are robust). It *is* asking a different kind of question (about policies, not models), preserving model diversity, and making disagreement visible and navigable. A caveat: policy robustness is evaluated relative to a specified decision criterion (welfare weights, risk aversion, constraints); AMTAIR targets epistemic pluralism across models, not the full space of normative disagreement about values and objectives.³⁵

³²Formats and losses: congressional testimony incentivizes binary framing through adversarial questioning (loses distributional nuance); advisory reports demand recommendations (loses disagreement structure); expert elicitation demands precise confidence intervals (loses ensemble reasoning); prediction markets produce one price (loses model identity).

³³Premature aggregation has often been the least bad option given infrastructure constraints. The critique here is of structural limitations that new technology and tools can relax.

³⁴What gets lost: model identity ("Bostrom says 35%, Christiano says 10%"), disagreement structure ("fast vs slow takeoff is key"), sensitivity ("if interpretability succeeds, estimate drops"), robustness ("all models agree uncoordinated deployment is risky"), assumption sources. Precisely the information needed for responsible pluralism.

³⁵Level-separation has precedents: Robust Decision Making (Lempert et al., 2006) explores policy performance across scenarios; info-gap theory (Ben-Haim, 2006) seeks robustness under deep uncertainty. The key distinction is structural: RDM treats scenarios as parameter variations over a shared model framework, while level-separation treats models as distinct causal theories that may disagree about *structure*, not just parameters. RDM asks "which policies are robust to parameter uncertainty?" Level-separation asks "which policies are robust across fundamentally different causal models?" The multi-functional queryability requirement — that the same model supports E, P, and I queries — is specific to the Pearl-based formulation here.

III.4 The Uncertainty Preservation Principle

Level-separation requires maintaining ensemble structure, but III.2 showed that current practices eliminate these structures. The remedy is clear: preserve uncertainty and information in its most structured form as long as possible.

Uncertainty Preservation Principle: Epistemic uncertainty and information should be maintained in its most structured form — distributions over models, not merely distributions over outcomes — and collapsed only when a specific decision requires a specific input.

An equivalent formulation: *minimize informational compression until decision-forcing moments*. Rather than framing this temporally ("aggregate late"), the principle is better understood informationally: request the least-compressed output your decision actually requires. Full distributions preserve maximum structure and information; confidence intervals preserve less; point estimates preserve less still; binary thresholds reduce everything to a yes or no. The question is always: what level of compression does this specific decision actually demand?³⁶

This principle itself follows from standard Bayesian reasoning about sufficient statistics and value of information. The contribution is empirical: governance institutions systematically violate it through the compression pathologies such as documented in III.2, and now infrastructure exists able to avoid this violation.

Why does this matter? Because uncertainty within and across models is information. When two models disagree, the disagreement reveals something — whether causal mechanisms are unknown, whether assumptions differ, whether scope varies. Putting the weighing aside, averaging Model A's $P(\text{catastrophe}) = 5\%$ with Model B's 40% to get 18% obscures structural uncertainty that differs fundamentally from "18% from one well-calibrated model." Moreover, different questions require different aggregations: expected damage calculations need means, worst-case planning needs tails. And preserved structure enables revision — if you aggregated early and Model B is later disconfirmed, you cannot cleanly remove its contribution. Compression is a one-way operation unless structure is maintained.

An operational protocol follows:

1. Maintain ensemble structures (distributions over models) as primary representation
2. Aggregate only when a specific decision requires a specific input
3. Aggregate appropriately to the question (mean for expected value, tail for worst-case)
4. Preserve underlying structure (DAGs) for subsequent queries
5. Make aggregation weights and methods explicit in decision justifications

This principle is *motivated* by what gets lost in compression (III.2) and *enables* what level-separation requires (III.3). The two strands converge here: understanding why current practices fail shows what must be preserved; understanding level-separation shows what the preservation is *for*.

III.5 Capability Requirements for Pluralism

Together, level-separation and uncertainty preservation imply specific capabilities. To ask "which policies are robust across models?" while maintaining ensemble structure, one must be able to: (1) *have multiple models to query across* → **Ensemble Maintenance**: models must persist as distinct, identifiable entities; (2) *ask each model about policy implications* → **Multi-Functional Queryability**: the *same* model must support explanation ("Why might this work?"), prediction ("What probability?"), and intervention ("What

³⁶This connects to the decision-theoretic literature on value of information (Howard, 1966): where information has value only insofar as it changes decisions. The uncertainty preservation principle inverts this insight: preserve structure until a specific decision determines the compression is necessary. Compare also to Savage (1954) on the relationship between probability, utility, and rational choice.

would it do?"); (3) *compare answers and aggregate when decisions require* → **Common Representation:** models must be comparable in a common format enabling structural comparison, with aggregation per-query, not permanent.

These requirements are *derived*, not posited: they follow from what level-separation and uncertainty preservation need to work. The question becomes: what representation satisfies all three?

III.6 A Workable Representation

Traditional approaches fail to satisfy all three requirements: the original MTAIR framework (Clarke et al., 2022) is multi-functional but requires prohibitive manual formalization effort per model; ensemble methods maintain models but collapse to outputs; argument mining extracts from text but provides no formal inference or representation.

CBNs represented as DAGs satisfy all three requirements. *Ensemble Maintenance:* Bayesian networks are objects, discrete structures with their own graphs and probabilities. A library naturally forms an ensemble and BMA can provide aggregation *when queries demand it*, but the underlying library persists. *Multi-Functional Queryability:* Within Pearl's (2009) structural causal model (SCM) framework, the same graph supports explanation (query causal structure), prediction (compute $P(Y|X)$), and intervention (apply do-calculus), given that the relevant causal assumptions hold. Crucially, do-calculus *defines* what intervention means: severing incoming arrows to the intervened node. AMTAIR produces candidate SCMs from arguments; interventional conclusions remain conditional on validation of those causal assumptions.³⁷ *Common Representation:* all models as Bayesian networks share a common format enabling structural comparison and principled aggregation.

But does requiring a common format impose a new form of monism — representational uniformity replacing theoretical uniformity? In principle, format-uniformity need not entail content-uniformity: models can disagree completely about causal structure, probability assignments, and mechanisms while sharing the same representational format. But whether this distinction holds under optimization pressure is genuinely uncertain. If an entire ecosystem optimized around Bayesian network extraction, the format could shape what gets modeled, marginalizing insights that resist probabilistic expression. This tension is not resolved here; it is an open question this work raises³⁸.

CBNs satisfy the requirements but traditionally require manual formalization; this is where AMTAIR contributes: extracting CBNs from natural language arguments automatically. Section IV demonstrates feasibility.

IV. AMTAIR: Demonstrating Feasibility

IV.1 System Architecture Overview

The requirements derived in Section III (ensemble maintenance, multi-functional queryability, common representation) are satisfied by DAGs in Pearl's representational framework in principle. AMTAIR demonstrates they can be satisfied in practice.

AMTAIR (Automated Modeling of Transformative AI Risks) makes model pluralism *visible* and actionable by extracting CBNs from natural language arguments.³⁹ Where expert reasoning exists as prose (implicit assumptions, unstated dependencies, qualitative hedges), AMTAIR extracts causal trees

³⁷Standard regression, decision tree, and neural network models do not natively support intervention reasoning, causal structure revelation, or mechanistic explanation without additional causal assumptions or structure.

³⁸Returned to in Section VI

³⁹Implementation details, code, and supplementary materials available at <https://vjmeyer.github.io/submission/>. See Meyer (2025a) for full technical documentation.

and formalizes structure that can be examined, compared, and queried. Where manual extraction required 20+ hours per model (Section I.3), AMTAIR reduces this to minutes plus focused validation.

The system processes arguments through two stages: *Stage 1* extracts argument structure (claims, support/attack relationships, hierarchical dependencies, parent/child nodes) into ArgDown format (building on Voigt, 2014; extended in Meyer, 2025a); *Stage 2* enriches structure with probabilistic information (priors, conditional probabilities, uncertainty ranges) into BayesDown⁴⁰ format (Meyer, 2025a).

IV.2 Why Formalization Matters

Why formalize natural language arguments into Bayesian networks? Formalization has multiple tangible benefits: (1) *making assumptions explicit*, as formalization reveals hidden premises driving disagreement; (2) *enabling systematic comparison*, as different vocabularies, arguments and models become structurally comparable; (3) *supporting robustness analysis*, testing which conclusions remain stable across models and sensitive to what conditions; and (4) *creating accountability*, as formal predictions can be tracked against reality.⁴¹

AMTAIR pursues *minimal necessary formalization*: the simplest structure to enable policy-level coordination, without claiming to capture everything models contain. Three constraints justify this: *tractability* (comprehensive formalization would require specifying thousands of relationships), *fidelity* (over-formalization creates false precision), and *epistemic humility* (models exceed any fixed representation).⁴² The threshold⁴³ is coordination-functional: sufficient formalization to compare policy implications, identify robust recommendations, make uncertainty explicit, and enable productive disagreement.

IV.3 The Extraction Pipeline

AMTAIR processes arguments through two stages. *Stage 1 (ArgDown)* extracts hierarchical causal structure (claims, support relationships, and dependencies) in human-readable format.⁴⁴ *Stage 2 (BayesDown)* adds probability information (priors, conditional probability tables, uncertainty ranges), producing complete Bayesian network specifications.⁴⁵ The separation enables targeted validation: structural errors differ from probability estimation errors.

AMTAIR has successfully extracted Carlsmith (2022)'s power-seeking AI argument (six conjunctive conditions with detailed sub-arguments, yielding a 23-node network)⁴⁶ and Bucknall & Dori-Hacohen (2022)'s near-term AI risk analysis (multi-path risk architecture mapping institutional pathways to existential outcomes). These extractions span argument types — from causal explanation *E* to sociotechnical risk architecture *I* — each preserving distinctive content while enabling cross-model comparison through shared formal structure.

Multi-Functional Queryability: Carlsmith Demonstration

⁴⁰See Appendix A for format specifications and implementation details.

⁴¹For assumptions: every conditional probability relationship must be specified, revealing hidden premises. For comparison: structural parallels and divergences become visible as graphs. For robustness: one can ask "Does conclusion C follow from models $M_1, M_2, \dots, M_n$?" For accountability: Bayesian networks document and track the specific conditional relationships which can be tracked over time.

⁴²As Mäki (2009) argues regarding economic models, unrealistic assumptions can serve legitimate purposes when they isolate mechanisms of interest; the question is whether the abstraction serves the user's goals, not whether it "matches reality."

⁴³This connects to Veit's anarchism challenge: minimal formalization resists the imperial ambition to capture everything in a single framework while acknowledging that coordination requires *some* common structure.

⁴⁴Example: For "When it rains, the grass gets wet. The sprinkler also makes the grass wet. However, when it rains, we usually don't run the sprinkler" the system extracts [Grass_Wet] + [Rain] + [Sprinkler] + [Rain] — where Rain influences both Grass_Wet directly and Sprinkler usage (nested + [Rain] under [Sprinkler]), producing a network with both a fork at Rain and a collider at Grass_Wet. See Appendix A for full specification.

⁴⁵Example: [Grass_Wet]: {"posteriors": {"p(wet|rain,sprinkler)": "0.99", "p(wet|rain,not-sprinkler)": "0.8", ...}} enables full Bayesian inference.

⁴⁶See Appendix A for the full ArgDown hierarchy and BayesDown probability tables.

The *same* extracted Carlsmith network⁴⁷ can serve all three epistemic functions:

- **Explanation (E):** Query causal structure — the 23-node graph reveals misaligned power-seeking as a critical hub where alignment difficulty, deployment pressures, and AI deception converge, with influence propagating through six causal levels to existential catastrophe.
- **Prediction (P):** Query probability distributions across the entire network — Bayesian inference yields $P(\text{catastrophe}) \approx 5\%$ across Carlsmith's six-premise conjunction given their probability estimates based on expert elicitations.
- **Intervention (I):** Apply do-calculus — intervening on deployment (mandatory safety certification) changes $P(\text{deployment}|\text{misaligned})$ from 0.6 to 0.1–0.2, reducing $P(\text{catastrophe})$ from $\sim 5\%$ to $\sim 1.0\text{--}1.9\%$. This differs from conditioning because interventions can also be modeled for example by severing or adding incoming causal arrows and nodes.

Calling Carlsmith's argument "explanatory" reflects its *primary emphasis*, not an exclusive category; the extracted network supports all three query modes. AMTAIR demonstrates that such multi-functional representations can be extracted from natural language.

IV.4 Policy Preference Innovation

One of AMTAIR's key innovations lies in mapping models to policy implications rather than forcing model selection. Models may be theoretically incommensurable, but their policy implications exist in common space. One can identify robust policies (supported across diverse models), contingent policies (dependent on specific models), and trade-off relationships (where models conflict).⁴⁸

BMA enters as a technique invoked when specific policy questions demand cross-model answers.⁴⁹ After answering the query, models remain distinct. A note on likelihoods: BMA requires $P(D|M)$ for each model. Once extracted into Bayesian networks, all models generate probability distributions by construction — this is what BayesDown achieves. The question is not whether explanatory or interventionist arguments *can* yield likelihoods (they can, once formalized), but how well-calibrated those estimates are compared to empirically grounded predictive models. Models optimized for specific purposes tend to outperform other models who in turn perform better for other purposes.

AMTAIR ensures faithfulness to the original sources by extracting the information multiple times, until convergence is observed. Furthermore, each model undergoes multi-stage validation (structural, probabilistic, semantic) and sensitivity analysis that reveals how much policy conclusions depend on less-certain estimates versus well-calibrated ones. The treatment is systematic rather than arbitrary: assumptions are explicit and revisable.⁵⁰

Concrete Example: Compute Governance

Consider chip export controls: scaling-focused models prioritize compute thresholds; alignment-focused models emphasize safety evaluations; coordination-focused models highlight verification mechanisms. AMTAIR identifies convergence (all three support mandatory safety evaluations for frontier models) while disagreements highlight areas requiring further research or negotiation.

When Robustness Analysis Yields No Robust Policies

⁴⁷For brevity, the carlsmith model is used to demonstrate the possibility of three different types of queries. Models specifically optimized for different E/P/I scientific purposes may perform even better.

⁴⁸Examples for policy domains include compute governance (thresholds, monitoring), international coordination (treaties, verification), research priorities (safety vs. capabilities), deployment regulations (testing, licensing) and so forth.

⁴⁹ $U(\text{policy}) = \sum_k P(M_k | D) \cdot U_k(\text{policy})$. Refer to Appendix D for methodology.

⁵⁰BMA is not monism: $P(\Delta | D)$ is a *distribution* that is carried forward, not a point estimate; aggregation is per-query not permanent; and the underlying ensemble grows rather than shrinks as more models are added. See Appendix B.

What if models recommend incompatible policies? Absence of robust policies is *informative*, not a failure; it surfaces genuine disagreement that demands attention rather than papering over it with premature consensus. The appropriate response depends on whether disagreement concerns facts (research to resolve), values (democratic deliberation), or mechanisms (scenario planning)⁵¹.

IV.5 Three Emphases, Each Multi-Functional

The Carlsmith demonstration shows multi-functional queryability within a single extraction. The principle generalizes across argument types with different primary emphases. Carlsmith (2022) emphasizes *why* alignment might fail (explanation). Bucknall & Dori-Hacohen (2022)'s risk architecture emphasizes *how* near-term AI amplifies existential risks through institutional pathways — state relations, corporate power, compromised political decision-making — where shared vulnerability nodes feed into multiple downstream risks simultaneously (intervention). Scaling laws (Branwen, 2020; Kaplan et al., 2020) emphasize *when* capabilities emerge (prediction). Each model has a distinctive emphasis, yet once extracted into Bayesian networks, each supports all three query modes by mathematical necessity but the "further away" a query is from a model's core strengths the higher the variance and lower information of the outputs. The Carlsmith and Bucknall extractions demonstrate how their DAG representations align with *E* and *I* respectively while enabling *P* queries not quite as well as dedicated quantitative prediction models.

IV.6 Validation and Quality

Automated extraction requires multi-layered validation. **Structural validation** checks that extracted argument graphs preserve logical relationships, ensuring that parent and child nodes correctly represent the original reasoning. **Probability validation** verifies that extracted estimates fall within reasonable ranges and conditional relationships satisfy basic coherence (probabilities sum to 1, conditionals do not contradict priors absurdly). **Semantic validation** confirms that claim labels accurately capture the original argument's meaning, not just syntactic patterns. Each validation can be performed probabilistically by repeatedly running the respective extraction processes and noticing convergences. Due to the mathematical attributes of the CBNs some forms of formal verification, especially coherence (as "Dutch-book proofness"), can be encoded and run deterministically as well.

AMTAIR's validation approach compares automated extractions against human expert extractions for benchmark arguments (Meyer, 2025a). On simple canonical examples (like rain-sprinkler-grass), automated extraction achieves high structural fidelity. On complex real-world cases like Carlsmith's power-seeking argument, accuracy varies more, particularly for long-range dependencies between distant claims. Probability estimates show higher variance than structural extraction. Rigorous validation with systematic accuracy metrics remains ongoing work; what the prototype demonstrates is *feasibility* of the extraction approach, not production-ready reliability.

Current validation requires 1-2 hours per model, a dramatic, approximate order of magnitude reduction from manual formalization effort. This shifts the value proposition to expanding feasible scope while maintaining quality through targeted oversight. The system achieves sufficient fidelity for decision support with transparent degradation patterns.⁵²

V. Critical Analysis: Limits and Challenges

AMTAIR has real limitations: computational, epistemological, and fundamental.⁵³

⁵¹See Appendix I for extended discussion of calibrated action under acknowledged uncertainty.

⁵²See Appendix E for more details on the validation methodology.

⁵³See Appendix I for extended technical discussion including anti-gaming mechanisms and systematic epistemic loss catalogs.

V.1 Computational Complexity

Extraction quality degrades with complexity: simple arguments achieve high fidelity; complex treatises come with higher variance. Scaling faces harsh realities: processing large document corpora requires substantial API costs and validation hours.⁵⁴

A technical concern: if individual extractions contain significant structural error, BMA aggregation may compound rather than correct bias. Ensemble methods can exhibit error-dampening, but this depends on errors being substantially uncorrelated across models — a condition that weakens precisely when models share theoretical foundations or training data.⁵⁵ The hierarchical grouping approach mitigates but does not eliminate this risk. Users should treat outputs as decision support requiring expert review, not ground truth.

The core challenge from computational complexity is inference across very large networks. As the number of nodes and edges increases inference costs tend to scale nonlinearly, in the worst case even exponentially. This constrains the maximal model sizes and complexity but given the progress in this area of computation (fundamentally, identical to neural networks used for today's LLMs) other bottlenecks and constraints such as human understanding and comprehension bind much earlier.

V.2 Epistemic Culture Clashes

AMTAIR's formalization inevitably privileges certain approaches and methodological cultures. Bayesian networks tend to favor quantitative over qualitative reasoning; hierarchical structure suits deductive over narrative arguments. Computer science, philosophy, and policy cultures have different epistemological approaches, with different views about what counts as known and to which degrees. The system captures the *predictive shadow* of each argument, what is expressible as probability distributions over causal structure — not necessarily all aspects or the full contents. Carlsmith's philosophical depth remains in the original; the extraction captures what is needed for policy coordination.

V.3 Automation Dependencies

Automation creates concerning dynamics: over-reliance on extractions, dependency on tools requiring expertise to validate, and gaming possibilities where arguments optimize for extraction rather than accuracy.⁵⁶ AMTAIR incorporates friction mechanisms: adversarial review, sensitivity scoring, validation proportional to weight, and structural complexity penalties. (Details in Appendix I.)

V.4 Fundamental Boundaries

Some limitations are inherent: AMTAIR cannot extract tacit knowledge embodied in practice, cannot resolve value disagreements (only clarify them), cannot escape decision-theoretic foundations,⁵⁷ cannot guarantee completeness, and cannot eliminate judgment. Not all theoretical frameworks are amenable to probabilistic formalization: approaches grounded in discourse analysis, phenomenology, or post-structuralist critique may resist it by their nature. For current AI governance, the most influential models

⁵⁴These constraints reflect current frontier model capabilities; rapid progress and collapsing costs suggest improved feasibility in coming years.

⁵⁵See Appendix B on model correlations

⁵⁶Buckingham Shum (2024) analyzes cognitive offloading risks with GenAI tools directly applicable here: users may accept formal models without critical engagement, and precise probabilities can create false authority masking extraction errors. Hendrycks (2023)'s analysis of evolutionary dynamics suggests gaming concerns may intensify as selection pressures favor AI systems producing formally compelling outputs regardless of underlying validity.

⁵⁷AMTAIR inherits assumptions from standard decision theory, treating choices as isolated actions evaluated by expected consequences. As Yudkowsky (2008) emphasized, fundamental problems in specifying goals correctly may precede questions of capability or alignment. For AI governance, where strategic interactions between agents implementing known decision procedures become increasingly relevant, AMTAIR's classical decision-theoretic foundations may require extension.

happen to come from quantitative traditions. But a system that claims to implement pluralism should be transparent about which epistemologies it systematically underserves — and about the risk that formalization incentives could, over time, marginalize precisely the qualitative insights that pluralism was supposed to protect.

V.5 Technology-Enabled Practice

AMTAIR became possible only with frontier LLMs (2023-2024). This is philosophically significant: the conceptual infrastructure developed here could be *articulated* before frontier LLMs but could not be *practiced* at scale. Earlier NLP could extract keywords and simple relations but lacked the semantic understanding to parse complex argumentative prose into causal structures with probability estimates. Frontier LLMs cross this threshold, making model pluralism operationalizable rather than merely philosophically defensible. The reflexive dimension (using AI to analyze AI risks) creates both capabilities and shared failure modes that deserve scrutiny.⁵⁸

VI. Conclusion

The Commerce official's challenge — "You have dozens of models pointing in different directions; how can we make policy based on your confusion?" — deserves a straight answer.

The confusion is real but mislocated. It stems from searching for "the best model", a question that Veit's analysis proves misleading for complex phenomena where different models serve different epistemic goals. A better question is: "How can we design policies that perform well across our models?" And *that* question is tractable.

This paper has tried to show how. One core move is level-separation: distinguishing model-level disagreements and uncertainties (which may be permanent and appropriate) from policy-level analysis (which may identify robust interventions despite those disagreements). What makes this possible is finding representations — Pearl's CBNs — that support different functions. Explanation, prediction, and intervention as different query modes over the same structures. Models formalized this way remain distinct, but their policy implications become comparable.

AMTAIR demonstrates that thanks to frontier LLMs, such formalizations can now be extracted from natural language arguments automatically and at scale. The philosophical significance depends on what one expects. For Veit's framework specifically, this tests the *practical* thesis (that multiplicity is necessary and tools can work with it), distinct from the analytical thesis about how philosophers should study science. I believe AMTAIR shows something modest but genuine: that frontier LLMs can help bridge the gap between Veit's theoretical insight (pluralism is permanent) and practical governance (decisions must be made).⁵⁹ If we ask the right questions at the right levels, we avoid collapsing pluralism into false consensus.

A harder question remains. Does model extraction into Pearl's framework impose a new form of monism, representational uniformity replacing theoretical uniformity? I think not, but the concern is serious. Pearl's framework does not demand that all models share the same causal structure or probability assignments; it demands only that they be expressible in a common format that supports multiple query modes. Models can disagree completely about causes, mechanisms, and likelihoods while still being comparable at the policy level and the dimensions we care about. The uniformity is in the *representation format*, not the *contents*. Whether this distinction holds under pressure, whether

⁵⁸See Appendix J for extended discussion of technology-enabled philosophy and theory-ladenness.

⁵⁹Future directions include prediction market integration, where logical induction (Garrabrant et al., 2016) formalizes Hayek's (1945) insight about markets as knowledge aggregators — extension to other domains (climate policy, biosecurity), and formal robustness metrics quantifying policy sensitivity to model weights.

format-monism inevitably distorts content-pluralism — for instance by systematically underrepresenting thick institutional narratives, Knightian uncertainty, or qualitative mechanisms that resist probabilistic expression — I cannot fully resolve here. It is an important open question this work raises.

What I am more confident about: current institutional practices fail. Congressional testimony, expert elicitation, advisory committees — all force the compression of ensemble knowledge into binary recommendations and false certainty. The information that responsible pluralism needs is lost long before it can be used. Even imperfect formalization that preserves model identity and enables policy-level queries is better than the status quo of model-as-rhetoric and premature aggregation.

The Commerce official example still deserves a concrete answer. Here it is: "We have multiple models because the world and our problems are genuinely complex. But we can still tell you this: *all* of these models support mandatory safety evaluations for frontier systems above certain compute thresholds, though they disagree on where exactly to set those thresholds. That disagreement matters and deserves further research. But the convergence on requiring such evaluations is robust; it holds across models. Act on the robust parts; invest in resolving the contested part. And when no consensus emerges between our models, we at least understand how uncertain the situation at hand really has to be."

That is what level-separation can look like in practice. Not algorithmic or forced consensus, but structured visibility into where models agree and disagree, enabling deliberation about which cruxes and uncertainties matter most for which decisions.

Veit established that different epistemic functions require different models. This paper has tried to show that permanent pluralism need not paralyze governance: that with the right infrastructure, model diversity becomes a resource for identifying robust action rather than an obstacle to it.⁶⁰ Whether AMTAIR or similar projects will prove adequate remains to be seen.⁶¹ But the possibility itself — actionable pluralism, not as practical burden but as methodology molded into epistemic infrastructure — seems worth pursuing.

This paper addresses one facet of a larger challenge. Companion work of this thesis explores complementary aspects: how the formalization process itself can help transform pre-paradigmatic discourse toward greater coherence (Meyer, 2025b),⁶² and how organizations can build genuine epistemic capacity for working with model ensembles and integrate live prediction market forecasts (Meyer, 2025c).⁶³ Together, these investigations address different dimensions of the same problem: enabling wise governance of transformative technologies rather than compressing hard-won knowledge and modeling into false certainty.

⁶⁰For AI governance specifically, AMTAIR can for example provide: systematic disagreement management, policy robustness analysis (Brundage et al., 2020), resource allocation guidance revealing which uncertainties matter most, forecasting infrastructure via prediction market integration, and institutional memory enabling learning from successes and failures.

⁶¹Three validation approaches would strengthen the framework: (1) coordination metrics comparing deliberation quality with/without AMTAIR; (2) user studies validating extractions against expert judgment; (3) longitudinal tracking of whether "robust across models" predicts policy success.

⁶²See Meyer (2025b) for analysis of how Kuhnian paradigm dynamics, cultural evolution of ideas, and abstract argumentation semantics can inform the transition from fragmented expert discourse to more structured deliberation.

⁶³See Meyer (2025c) for analysis of hierarchical Bayesian frameworks, prediction market integration with contamination prevention, and institutional design for maintaining epistemic diversity.

References

- Ben-Haim, Y. (2006). *Info-Gap Decision Theory: Decisions Under Severe Uncertainty* (2nd ed.). Academic Press.
- Berger, J. O. (2006). The Case for Objective Bayesian Analysis. *Bayesian Analysis*, 1(3), 385–402. <https://doi.org/10.1214/06-BA115>
- Branwen, G. (2020, May 28). *The Scaling Hypothesis*. Gwern.net. <https://gwern.net/scaling-hypothesis>
- Brundage, M., Avin, S., Wang, J., Belfield, H., Krueger, G., Hadfield, G., Khlaaf, H., Yang, J., Toner, H., Fong, R., Maharaj, T., Koh, P. W., Hooker, S., Leung, J., & Dafoe, A. (2020). Toward Trustworthy AI Development: Mechanisms for Supporting Verifiable Claims. *arXiv Preprint*. <https://doi.org/10.48550/arXiv.2004.07213>
- Buckingham Shum, S. (2024). Generative AI for Critical Analysis: Practical Tools, Cognitive Offloading and Human Agency. *Joint Proceedings of LAK 2024 Workshops*, 205–213. <https://ceur-ws.org/Vol-3667/>
- Bucknall, B. S., & Dori-Hacohen, S. (2022). Current and Near-Term AI as a Potential Existential Risk Factor. *Proceedings of the 2022 AAAI/ACM Conference on AI, Ethics, and Society*, 119–129. <https://doi.org/10.1145/3514094.3534146>
- Carlsmith, J. (2022). *Is power-seeking AI an existential risk?* <https://arxiv.org/abs/2206.13353>
- Cartwright, N. (1999). *The Dappled World: A Study of the Boundaries of Science*. Cambridge University Press.
- Christiano, P. (2018, April 7). *Clarifying AI Alignment*. AI Alignment Forum. <https://www.alignmentforum.org/posts/ZeE7EKHTFMBs8eMxn/clarifying-ai-alignment>
- Clarke, S., Cottier, B., Englander, A., Eth, D., Manheim, D., Martin, S. D., & Rice, I. (2022). *Modeling Transformative AI Risks (MTAIR) Project – Summary Report*. <https://doi.org/10.48550/ARXIV.2206.09360>
- Dafoe, A. (2018). AI governance: A research agenda. *Governance of AI Program, Future of Humanity Institute, University of Oxford: Oxford, UK*, 1442, 1443. <https://www.fhi.ox.ac.uk/wp-content/uploads/GovAI-Agenda.pdf>
- Duhem, P. M. M. (1954). *The Aim and Structure of Physical Theory*. Princeton University Press.
- Dung, P. M. (1995). On the Acceptability of Arguments and its Fundamental Role in Nonmonotonic Reasoning, Logic Programming and n-Person Games. *Artificial Intelligence*, 77, 321–357. [https://doi.org/10.1016/0004-3702\(94\)00041-X](https://doi.org/10.1016/0004-3702(94)00041-X)
- Garrabrant, S., Benson-Tilsen, T., Critch, A., Soares, N., & Taylor, J. (2016, September 12). *Logical Induction*. <https://arxiv.org/abs/1609.03543>
- Gigerenzer, G., Todd, P. M., & ABC Research Group. (1999). *Simple Heuristics That Make Us Smart*. Oxford University Press.
- Grüne-Yanoff, T., & Marchionni, C. (2018). Modeling Model Selection in Model Pluralism. *Journal of Economic Methodology*, 25(3), 265–275. <https://doi.org/10.1080/1350178X.2018.1488572>
- Hansen, P. R., Lunde, A., & Nason, J. M. (2011). The Model Confidence Set. *Econometrica*, 79(2), 453–497. <https://doi.org/10.3982/ECTA5771>
- Hayek, F. A. (1945). The Use of Knowledge in Society. *The American Economic Review*, 35(4), 519–530. <https://www.jstor.org/stable/1809376>
- Heesen, R., Bright, L. K., & Zucker, A. (2019). Vindicating Methodological Triangulation. *Synthese*, 196(8), 3067–3081. <https://doi.org/10.1007/s11229-016-1294-7>
- Hendrycks, D. (2023). *Natural Selection Favors AIs over Humans*. <https://arxiv.org/abs/2303.16200>

- Hoeting, J. A., Madigan, D., Raftery, A. E., & Volinsky, C. T. (1999). Bayesian Model Averaging: A Tutorial. *Statistical Science*, 14(4), 382–417. <https://doi.org/10.1214/ss/1009212519>
- Howard, R. A. (1966). Information Value Theory. *IEEE Transactions on Systems Science and Cybernetics*, 2(1), 22–26. <https://doi.org/10.1109/TSSC.1966.300074>
- Hubinger, E., van Merwijk, C., Mikulik, V., Skalse, J., & Garrabrant, S. (2019). *Risks from Learned Optimization in Advanced Machine Learning Systems*. <https://doi.org/10.48550/arXiv.1906.01820>
- Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., Gray, S., Radford, A., Wu, J., & Amodei, D. (2020). Scaling laws for neural language models. *arXiv Preprint arXiv:2001.08361*. <https://doi.org/10.48550/arXiv.2001.08361>
- Kuhn, T. (1962). The structure of scientific revolutions. *International Encyclopedia of Unified Science*, 2(2).
- Lempert, R. J., Groves, D. G., Popper, S. W., & Bankes, S. C. (2006). A General, Analytic Method for Generating Robust Strategies and Narrative Scenarios. *Management Science*, 52(4), 514–528. <https://doi.org/10.1287/mnsc.1050.0472>
- Levins, R. (1966). The Strategy of Model Building in Population Biology. *American Scientist*, 54(4), 421–431.
- Mäki, U. (2009). Realistic Realism about Unrealistic Models. In D. Ross & H. Kincaid (Eds.), *The Oxford Handbook of Philosophy of Economics* (pp. 68–98). Oxford University Press.
- Meyer, V. J. (2025a). *Automating Transformative AI Risk Modeling* [Forthcoming, University of Bayreuth]. <https://vjmeyer.github.io/submission/>
- Meyer, V. J. (2025b). *From pre-paradigmatic chaos to constructive science: How AMTAIR could transform AI governance through evolutionary intervention* [Forthcoming]. University of Bayreuth. https://vjmeyer.github.io/submission/papers/2025-VJM_ParadigmFormation-AMTAIR_EvoRatPaper.html
- Meyer, V. J. (2025c). *Live models: Integrating prediction markets into AMTAIR’s bayesian risk networks: Managing contamination loops in collaborative forecasting systems* [Forthcoming]. University of Bayreuth. https://vjmeyer.github.io/submission/papers/2025-VJM_PredictionMarketIntegration-AMTAIR_EpiOrgsPaper.html
- Meyer, V. J. (2025d). *Practicing model pluralism: From expert intuitions to governance infrastructure* [Forthcoming]. University of Bayreuth. https://vjmeyer.github.io/submission/papers/2025-VJM_ModelPluralism-AMTAIR_PhilOfEconPaper.html
- OpenAI. (2023). *GPT-4 Technical Report*. OpenAI. <https://arxiv.org/abs/2303.08774>
- Pearl, J. (2009). *Causality: Models, reasoning and inference* (2nd ed.). Cambridge University Press.
- Prokudin, D. E., Lisanyuk, E. N., & Baymurov, I. R. (2024). Visualization Functions in Argumentation Representation Software. *Scientific Visualization*, 16(3). <https://sv-journal.org/2024-3/11/en.pdf>
- Quine, W. V. Orman. (1951). Two Dogmas of Empiricism. *Philosophical Review*, 60(1), 20–43.
- Rodrik, D. (2015). *Economics Rules: The Rights and Wrongs of the Dismal Science*. W. W. Norton.
- Savage, L. J. (1954). *The Foundations of Statistics*. John Wiley & Sons.
- Stanford, P. K. (2006). *Exceeding Our Grasp: Science, History, and the Problem of Unconceived Alternatives*. Oxford University Press. <https://doi.org/10.1093/0195174089.001.0001>
- Veit, W. (2019). Model Pluralism. *Philosophy of the Social Sciences*, 50(2), 91–114. <https://doi.org/10.1177/0048393119894897>
- Veit, W. (2020). Model Anarchism. *THEORIA. An International Journal for Theory, History and Foundations of Science*, 38(2), 225–245. <https://doi.org/10.1387/theoria.23849>
- Veit, W. (2021). Model Diversity and the Embarrassment of Riches. *Journal of Economic Methodology*, 28(3), 291–303. <https://doi.org/10.1080/1350178X.2021.1898660>

- Veit, W., & Browning, H. (2023). Neural Networks, AI, and the Goals of Modeling. *Behavioral and Brain Sciences*, 46, e411. <https://doi.org/10.1017/S0140525X23001681>
- Villani, C. (2009). *Optimal Transport: Old and New*. Springer. <https://doi.org/10.1007/978-3-540-71050-9>
- Voigt, C. (2014). Argdown and the Stacked Masonry Layout: Two User Interfaces for Non-Expert Users. *Computational Models of Argument: Proceedings of COMMA 2014*, 483–484. <https://doi.org/10.3233/978-1-61499-436-7-483>
- Yao, Y., Vehtari, A., Simpson, D., & Gelman, A. (2018). Using Stacking to Average Bayesian Predictive Distributions (with Discussion). *Bayesian Analysis*, 13(3), 917–1007. <https://doi.org/10.1214/17-BA1091>
- Yudkowsky, E. (2008). Artificial Intelligence as a positive and negative factor in global risk. In E. Yudkowsky, *Global Catastrophic Risks*. Oxford University Press. <https://doi.org/10.1093/oso/9780198570509.003.0021>
- Zollman, K. J. S. (2010). The Epistemic Benefit of Transient Diversity. *Erkenntnis*, 72(1), 17–35. <https://doi.org/10.1007/s10670-009-9194-6>
-

Appendices

Cross-Paper Integration Note: This paper (Meyer, 2025d) can be read independently. It is part of a three-paper thesis on AMTAIR. Companion papers address: (1) pre-paradigmatic to paradigmatic discourse transformation through formalization (Meyer, 2025b); and (2) prediction market integration with contamination prevention (Meyer, 2025c). All papers are available at <https://vjmeyer.github.io/submission/papers/> and at <https://www.amtair.org/publications/papers/>. Cross-references throughout indicate where content is developed further in companion papers.

Appendix A: ArgDown and BayesDown Specification

The following specifications describe the formats developed for AMTAIR (Meyer, 2025a). For related formal argumentation theory (Dung's Abstract Argumentation Frameworks), see Appendix H.

ArgDown Syntax

ArgDown (Voigt, 2014) provides markdown-inspired notation for hierarchical argument structure. Basic elements:

```
[ClaimID]: Claim text
+ [SupportID]: Supporting claim
- [AttackID]: Attacking claim
[ClaimA] -> [ClaimB]: A attacks B
```

Extended syntax supports nested argumentation, cross-references between claims, grouping, and metadata (tags, sources, confidence levels). Square brackets denote node identifiers; +/- indicate support/attack; indentation shows hierarchy.

Example from AI safety:

```
[AGIRisk]: Advanced AI poses existential risk
+ [PowerSeeking]: AI systems will pursue power instrumentally
+ [Instrumental]: Power is convergent instrumental goal
+ [Bostrom2014]: Bostrom's orthogonality thesis
- [AlignmentSuccess]: Alignment research will succeed
+ [Capability]: Systems will have sufficient capability
+ [ScalingLaws]: Capability scales with compute/data
- [Deployment]: Deployment will be controlled
```

BayesDown Extension

BayesDown (Meyer, 2025a) adds JSON probability metadata to ArgDown structure:

```
[NodeID]: Description.
{"instantiations": ["state1", "state2"],
 "priors": {"p(state1)": "0.7", "p(state2)": "0.3"},
 "posteriors": {
   "p(state1|parent1,parent2)": "0.9",
   "p(state1|parent1,not-parent2)": "0.5",
   "p(state1|not-parent1,parent2)": "0.6",
```

```
"p(state1|not-parent1,not-parent2)": "0.1"
}}
```

The extraction process identifies node states, extracts prior and conditional probabilities for each parent combination, and verifies coherence. The resulting BayesDown specifications convert directly to Bayesian networks: nodes become random variables, edges represent conditional dependencies, and standard inference algorithms apply.

Example Extraction: Carlsmith (2022)

Full extraction of Carlsmith (2022) "Is Power-Seeking AI an Existential Risk?" demonstrates BayesDown in practice. Six conjunctive conditions each with probability ranges:

1. $P(\text{timely development}) = 0.65$ [0.40-0.90]: "By 2070, it will become possible and financially feasible to build relevantly powerful and agentic AI systems"
2. $P(\text{power-seeking}|\text{development}) = 0.60$ [0.40-0.80]: "Some of these systems will be power-seeking"
3. $P(\text{sufficient capability}|\text{power-seeking,development}) = 0.50$ [0.30-0.70]: "There will be a 'crunch time' where systems are capable enough that their power-seeking behavior could lead to existential catastrophe"
4. $P(\text{misalignment}|\text{capability,power-seeking,development}) = 0.70$ [0.50-0.90]: "This power-seeking will in fact be misaligned power-seeking"
5. $P(\text{deployment}|\text{misalignment,capability,power-seeking,development}) = 0.60$ [0.40-0.80]: "Deployed systems will be misaligned in aggregate"
6. $P(\text{catastrophe}|\text{deployment,misalignment,capability,power-seeking,development}) = 0.70$ [0.40-0.90]: "This will lead to existential catastrophe"

Overall $P(\text{catastrophe}) = \text{Product of conditionals} \approx 0.05\text{-}0.10$ (5-10%)

Network structure: Sequential dependencies where later nodes conditional on earlier. Enables sensitivity analysis: Which uncertainty matters most? Answer: Development and power-seeking drive bulk of variation.

Full ArgDown Extraction

The complete hierarchical extraction reveals the 23-node causal structure underlying Carlsmith's argument.⁶⁴

```
[Existential_Catastrophe]: The destruction of humanity's long-term
potential due to AI systems we've lost control over.
- [Human_Disempowerment]: Permanent and collective disempowerment
of humanity relative to AI systems.
- [Scale_Of_Power_Seeking]: Power-seeking scaling to permanently
disempowering all of humanity.
- [Misaligned_Power_Seeking]: Deployed AI systems seeking
power in unintended and high-impact ways.
- [APS_Systems]: AI systems with advanced capabilities,
agentic planning, and strategic awareness.
```

⁶⁴Instantiation states (e.g., TRUE/FALSE, DEPLOY/WITHHOLD) and full metadata are specified in the complete extraction; see Meyer (2025a) and the public repository for the full BayesDown specification.

- [Advanced_AI_Capability]: Systems outperforming humans on tasks granting significant power.
 - [Agentic_Planning]: Systems making and executing plans based on world models.
 - [Strategic_Awareness]: Systems with models accurately representing power dynamics.
 - [Difficulty_Of_Alignment]: Harder to build aligned than misaligned systems attractive to deploy.
 - [Instrumental_Convergence]: Misaligned objectives tend to seek power instrumentally.
 - [Problems_With_Proxies]: Proxy objectives break correlations with intended goals.
 - [Problems_With_Search]: Search processes can yield systems with different objectives than intended.
 - [Deployment_Decisions]: Decisions to deploy potentially misaligned AI systems.
 - [Incentives_To_Build_APS]: Strong incentives to build and deploy APS systems.
 - [Usefulness_Of_APS]: APS systems useful for many valuable tasks.
 - [Competitive_Dynamics]: Competitive pressures between AI developers.
 - [Deception_By_AI]: AI systems deceiving humans about their true objectives.
 - [Corrective_Feedback]: Human society implementing corrections after observing problems.
 - [Warning_Shots]: Observable failures in weaker systems before catastrophic risks.
 - [Rapid_Capability_Escalation]: Capabilities escalating very rapidly, limiting correction time.
- [Barriers_To_Understanding]: Difficulty understanding advanced AI internals.
- [Misaligned_Power_Seeking]
- [Adversarial_Dynamics]: Adversarial human-AI relationships.
- [Misaligned_Power_Seeking]
- [Stakes_Of_Error]: Escalating impact of mistakes.
- [Misaligned_Power_Seeking]

The main causal chain contains 20 nodes; three auxiliary nodes (Barriers_To_Understanding, Adversarial_Dynamics, Stakes_Of_Error) are modeled as downstream consequences of Misaligned_Power_Seeking. Maximum depth is six causal levels. The hub node Misaligned_Power_Seeking receives input from three parent branches (APS_Systems, Difficulty_Of_Alignment, Deployment_Decisions) and feeds into the three auxiliary nodes, making it the most structurally significant node — consistent with Carlsmith's argument that misaligned power-seeking is the critical crux.

BayesDown Probability Tables (Excerpt)

Three representative nodes illustrate increasing conditional probability table (CPT) complexity. A leaf node carries only prior probabilities:

```
[Advanced_AI_Capability]: Systems outperforming humans on tasks
granting significant power.
{"instantiations": ["TRUE", "FALSE"],
 "priors": {"p(TRUE)": "0.80", "p(FALSE)": "0.20"}}
```

A two-parent node specifies probabilities conditional on each parent combination:

```
[Corrective_Feedback]: Human society implementing corrections.
{"instantiations": ["EFFECTIVE", "INEFFECTIVE"],
 "priors": {"p(EFFECTIVE)": "0.60"},
 "posteriors": {
  "p(EFFECTIVE|warning=OBS, escalation=T)": "0.40",
  "p(EFFECTIVE|warning=OBS, escalation=F)": "0.80",
  "p(EFFECTIVE|warning=UNOBS, escalation=T)": "0.15",
  "p(EFFECTIVE|warning=UNOBS, escalation=F)": "0.50"}}
```

The hub node requires eight conditional entries for its three parents:

```
[Misaligned_Power_Seeking]: Deployed AI systems seeking power in
unintended and high-impact ways.
{"instantiations": ["TRUE", "FALSE"],
 "priors": {"p(TRUE)": "0.338"},
 "posteriors": {
  "p(TRUE|aps=T, align_hard=T, deploy=DEPLOY)": "0.90",
  "p(TRUE|aps=T, align_hard=T, deploy=WITHHOLD)": "0.10",
  "p(TRUE|aps=T, align_hard=F, deploy=DEPLOY)": "0.25",
  "p(TRUE|aps=T, align_hard=F, deploy=WITHHOLD)": "0.05",
  "p(TRUE|aps=F, align_hard=T, deploy=DEPLOY)": "0.0",
  "p(TRUE|aps=F, align_hard=T, deploy=WITHHOLD)": "0.0",
  "p(TRUE|aps=F, align_hard=F, deploy=DEPLOY)": "0.0",
  "p(TRUE|aps=F, align_hard=F, deploy=WITHHOLD)": "0.0"}}
```

The CPTs reveal structural insights invisible in the original prose. When APS systems do not exist ($\text{aps}=\text{F}$), $P(\text{misaligned_power_seeking}) = 0.0$ regardless of alignment difficulty or deployment decisions — advanced capabilities are a necessary precondition. When all three risk factors align ($\text{aps}=\text{T}$, $\text{align_hard}=\text{T}$, $\text{deploy}=\text{DEPLOY}$), $P = 0.90$. For corrective feedback, observed warning shots without rapid escalation yield $P(\text{effective}) = 0.80$, but unobserved warning shots with rapid escalation drops effectiveness to 0.15. These conditional relationships are what enable the do-calculus interventions demonstrated in Section IV.3.

Network Topology

The complete extraction yields 23 nodes across six causal levels. Eleven root nodes carry only prior probabilities (e.g., $\text{Advanced_AI_Capability}$: $p(\text{TRUE}) = 0.80$). Twelve nodes carry conditional probability tables specifying how parent state combinations determine child state probabilities. The hub node $\text{Misaligned_Power_Seeking}$ has in-degree 3 (from APS_Systems , $\text{Difficulty_Of_Alignment}$,

Deployment_Decisions), making it the network's most structurally complex node. Three auxiliary nodes (Barriers_To_Understanding, Adversarial_Dynamics, Stakes_Of_Error) are modeled as downstream consequences conditioned on Misaligned_Power_Seeking, capturing factors that co-occur with and amplify the central risk without belonging to the primary causal chain.

Implementation Pipeline

Prototype uses:

- **LLM:** Frontier language models for extraction
- **Prompting:** Multi-stage prompts with examples
- **Validation:** Human review with source linking
- **Storage:** PostgreSQL database with version control
- **Visualization:** D3.js for interactive networks
- **Analysis:** Python (pgmpy library) for Bayesian inference

Computational Requirements: Extraction takes ~5-15 minutes per model; validation requires ~1-2 hours of expert review. Accuracy varies with argument complexity; human validation is recommended for high-stakes applications. The approach scales feasibly to dozens of models.

Full implementation available: https://github.com/SingularitySmith/AMTAIR_Prototype

Appendix B: BMA Mathematics Extended

Formal BMA Framework

Given data D , model set $\mathcal{M} = \{M_1, M_2, \dots, M_K\}$, and quantity of interest Δ :

Posterior model probability: $P(M_j | D) = P(D | M_j) \cdot P(M_j) / P(D)$

Where: - $P(D | M_j)$ is marginal likelihood (integrated likelihood) - $P(M_j)$ is prior model probability - $P(D) = \sum_j P(D | M_j) \cdot P(M_j)$ (normalizing constant)

Marginal likelihood: $P(D | M_j) = \int P(D | \theta_j, M_j) \cdot P(\theta_j | M_j) d\theta_j$

BMA prediction: $P(\Delta | D) = \sum_j P(\Delta | M_j, D) \cdot P(M_j | D)$

Accounting for Model Correlations

When models share assumptions or data sources, treating them as independent inflates apparent consensus. Hierarchical BMA partitions model space $\mathcal{M}$ into families $\mathcal{F} = \{F_1, F_2, \dots, F_m\}$:

$$P(\Delta | D) = \sum_i P(\Delta | F_i, D) \cdot P(F_i | D)$$

This two-stage approach prevents over-counting when models cluster.

Practical Implementation for AMTAIR

1. Identify model families (e.g., "alignment pessimist cluster," "scaling optimist cluster")
2. Apply hierarchical BMA with family-level and within-family weights
3. Use Model Confidence Sets to identify core high-confidence subset
4. Employ forest methods when exact inference computationally prohibitive

Forecasting with Incomplete Models

A deeper challenge confronts standard Bayesian approaches: they typically require complete model specification: prior distributions over all relevant variables, conditional probability tables for all relationships. But genuine model uncertainty often means one does not even know the full structure, let alone precise probabilities.

Logical induction (Garra-brant et al., 2016) offers a partial response: it demonstrates that computationally bounded agents can maintain coherent beliefs even with incomplete models, converging toward consistency over time without requiring complete specification upfront. For AMTAIR, this provides theoretical grounding for relaxing BMA's completeness requirements — extracted models capture partial rather than total structure, yet principled combination remains possible because the ensemble need not be fully specified to yield useful policy-level outputs. The practical implication: one can work with what is available, treating incompleteness as a feature of the epistemic situation rather than a defect to be eliminated before analysis can proceed.

Handling Model Correlations

A critical technical issue often overlooked: models in the ensemble are rarely independent. When models share theoretical foundations, draw on similar data, or emerge from connected research communities, their errors correlate. Standard BMA assumes independence, but this can dramatically overstate confidence when models fail together.

AMTAIR addresses correlation through several mechanisms. First, during extraction, the system tracks model provenance — which assumptions, data sources, and theoretical commitments each model shares. Second, correlation structures can be explicitly modeled using hierarchical Bayesian approaches, where models sharing foundations are grouped (see "Hierarchical structure for correlated models" below). Third, sensitivity analysis across alternative weight specifications reveals how much conclusions depend on independence assumptions, enabling conservative estimates when correlations are suspected but not precisely known.

Statistical Implementation Methods

The BMA equation is conceptually straightforward, but practical implementation must handle two challenges: computing marginal likelihoods (the integral over parameter space) and deciding which models belong in the ensemble at all.

For the first, AMTAIR can employ approximation methods when exact integration is infeasible — a common situation for complex Bayesian networks with many nodes.

For the second, Model Confidence Set (MCS) procedures (Hansen et al., 2011) provide a principled approach: rather than selecting a single "best" model, MCS identifies the set of models that cannot be statistically distinguished given available data. This aligns with philosophical pluralism: models are retained not because they are all "true" but because the evidence cannot yet discriminate among them. The MCS approach makes the connection between statistical practice and epistemic humility concrete — the size of the confidence set directly reflects how much the data constrains model choice.

The Prior Assignment Challenge

A fundamental limitation deserves explicit acknowledgment: BMA requires prior model probabilities $P(M_k)$, yet this is precisely where the embarrassment of riches bites hardest. If principled methods for assigning model priors existed, one might not face an embarrassment of riches in the first place. AMTAIR's contribution is not to solve this problem but to relocate it: from model selection (intractable under underdetermination) to prior elicitation (difficult but more tractable through methods like expert aggregation, track record analysis, or reference priors). The selection problem persists but becomes amenable to systematic treatment rather than arbitrary choice.

Toward a Methodology for Prior Assignment

Three design principles guide AMTAIR's approach to prior assignment, aiming to make judgment transparent and revisable rather than eliminating it:

Reference priors tied to model quality. Following Berger (2006)'s objective Bayesian approach, priors can be derived from structural properties of models: how fully they specify causal pathways, whether

their probability assignments are internally consistent (free of Dutch book vulnerability), and — where testable predictions exist, how well-calibrated their stated confidences have been historically.

Hierarchical structure for correlated models. When models share theoretical foundations (for example, multiple scaling-based forecasts from the same research tradition), their errors correlate. Standard BMA assumes independence, overstating confidence when correlated models agree. Grouping models by provenance and operating BMA at two levels (within and across groups) prevents "ballot stuffing" where similar worldviews dominate through sheer numbers.

Sensitivity analysis as validation. Most importantly, reporting how conclusions change under alternative prior specifications transforms prior assignment from hidden assumption to explicit parameter. Decision-makers can assess whether their prior judgments fall within robustness bounds, treating the formal analysis as conditional rather than definitive.

Coherence Under Computational Constraints

Traditional Dutch book arguments justify Bayesian probability: agents with incoherent beliefs can be exploited through sure-loss betting sequences. But these arguments assume logical omniscience, i.e., agents who instantly recognize all logical implications. Real agents, operating under computational constraints, cannot achieve such coherence.

Recent extensions address this gap: logical induction (Garraabrant et al., 2016) satisfies coherence constraints appropriate for bounded agents — no polynomial-time adversary can extract unbounded profit from a logical inductor. This matters for AMTAIR because the models being combined are themselves products of bounded reasoning. The ensemble inherits computational limitations from its components. Knowing that appropriate coherence constraints exist even for computationally bounded systems provides theoretical grounding for imperfect-but-principled aggregation. This is not approximating ideal rationality; it is implementing a distinct form of bounded rationality with its own coherence properties.

Appendix C: AMTAIR Technical Implementation

Technical architecture and computational requirements are covered in Appendix A (Implementation Pipeline). For full implementation details, source code, and documentation, see: <https://vjmeyer.github.io/submission/>

Appendix D: Policy Preference Mapping

See also: Meyer (2025b) (Appendix D) for extended mathematical framework including model selection vs. model averaging vs. robust policy approaches.

Mapping Models to Policy Implications

Step 1 - Policy space definition: Identify actionable interventions, define policy dimensions (compute governance, research funding, international coordination), create policy vector representation.

Step 2 - Model-to-policy elicitation (semi-automated): For each model, extract recommendations through explicit text analysis and counterfactual reasoning (automated), supplemented by expert elicitation for implicit policy implications (manual). This step requires approximately 30 minutes of human oversight per model.

Step 3 - Robustness analysis: Identify policies recommended by multiple models, calculate agreement scores, weight by model posteriors.

Step 4 - Uncertainty quantification: Sensitivity analysis across weight schemes, contingency identification for contested assumptions.

Example: Compute Governance Policy Vectors

Table 1: Illustrative policy vector scores (1-10 scale) across arguments with different emphases for compute governance interventions. Scores are derived from ad-hoc qualitative assessment of each argument's policy implications, not from automated AMTAIR output. Column labels indicate primary emphasis.

Dimension	Carlsmith (E-emphasis)	Scaling (P-emphasis)	Coordination (I-emphasis)
Export controls	7	9	5
Domestic threshold	8	5	6
International monitoring	6	8	10
Safety research	9	4	7

Robustness finding: All models support substantial international monitoring (high agreement). Models diverge on export controls and safety research (model-dependent).

Appendix E: Validation Methodology

See also: Meyer (2025c) for contamination prevention methodology when integrating external forecast data — particularly the hierarchical Bayesian framework (Appendix A) and cascade detection metrics.

Multi-Pass Extraction Protocol

1. Initial extraction with temperature = 0.7
2. Three additional extractions with different seeds
3. Comparison: structural consistency, relationship agreement, probability coherence
4. Flag discrepancies for human review
5. Consensus extraction when consistency high; manual resolution when low

Expert Review Procedures

Phase 1 - Structural validation: Does ArgDown capture argument correctly? Are claims properly identified? Are relationships accurate?

Phase 2 - Probability validation: Are numerical estimates reasonable? Do conditional probabilities match argument? Are probabilities internally coherent?

Phase 3 - Completeness check: Are important considerations omitted? Are there alternative structures?

Consistency Checking Algorithms

Automated checks: - Probability axioms: All probabilities $0 \leq p \leq 1$ - Normalization: Complementary probabilities sum to 1 - Conditional table completeness: All parent combinations specified - DAG property: No cycles in causal graph - Reachability: All nodes connected to argument structure

Quality Metrics

- Structural accuracy: % of nodes/edges correct
- Probability calibration: Agreement with expert estimates
- Completeness: % of argument content captured
- Consistency score: Agreement across multiple extractions

Appendix F: Abbreviations

Table 2: List of abbreviations used in this paper.

Abbreviation	Full Term
AAF	Abstract Argumentation Framework
AI	Artificial Intelligence
AMTAIR	Automated Modeling of Transformative AI Risks
ArgDown	Argumentation Markup Language (structure extraction format)
BayesDown	Probabilistic extension of ArgDown (adds distributions)
BMA	Bayesian Model Averaging
CBN	Causal Bayesian Network
CPT	Conditional Probability Table
DAG	Directed Acyclic Graph
E/P/I	Explanatory/Predictive/Interventionist (epistemic functions)
LLM	Large Language Model
MCS	Model Confidence Set
MTAIR	Modeling of Transformative AI Risks [manual predecessor to AMTAIR; Clarke et al. (2022)]
NLP	Natural Language Processing
RDM	Robust Decision Making
RLHF	Reinforcement Learning from Human Feedback

Appendix G: Glossary

Table 3: Key terms used in this paper.

Term	Definition	Source
Bayesian Model Averaging (BMA)	A statistical method that combines predictions from multiple models by weighting each model's contribution according to its posterior probability given observed data, rather than selecting a single "best" model.	Hoeting et al. (1999)

Term	Definition	Source
Causal Bayesian Network (CBN)	A directed acyclic graph in which nodes represent variables and edges represent causal relationships, equipped with conditional probability tables that quantify the strength of each causal link. CBNs support explanation (reading causal structure), prediction (computing probability distributions), and intervention (applying do-calculus to simulate policy changes).	Pearl (2009)
Embarrassment of Riches	The challenge arising when multiple models adequately explain observed data, making principled selection among them difficult without additional criteria.	Grüne-Yanoff & Marchionni (2018)
Frontier Model	A large language model at or near the current capability frontier, characterized by strong performance on complex reasoning, natural language understanding, and generalization across domains.	OpenAI (2023)
Large Language Model (LLM)	A neural network trained on large text corpora to predict and generate text, capable of complex reasoning, summarization, and — crucially for AMTAIR — extracting structured representations from unstructured arguments.	OpenAI (2023)
Level-Separation	The thesis that model-level pluralism is compatible with policy-level convergence: one can ask "which policies are robust across models?" without first resolving "which model is the best?" Distinguishes the level at which convergence is sought.	This paper, III.3
Model Pluralism	The philosophical position that scientific understanding requires maintaining multiple models serving different epistemic functions, rather than seeking convergence on a single unified model.	Veit (2019)
Multi-Functional Queryability	The property of a representation that supports multiple epistemic functions — explanation, prediction, and intervention — through different query modes on the same underlying structure. CBNs possess this property.	This paper, III.5; Pearl (2009)

Term	Definition	Source
Posterior Probability	In Bayesian statistics, the updated probability of a hypothesis after observing evidence, calculated by combining prior probability with the likelihood of the evidence given the hypothesis.	Stanford Encyclopedia of Philosophy
Predictive Shadow	What BayesDown extraction captures of a model: the aspects expressible as probability distributions over causal structure. Extraction preserves multi-functional queryability but loses the full content of the original argument (philosophical depth, rhetorical force, contextual nuance).	This paper, II.4; see Appendix I for systematic catalog of epistemic losses
Tacit Ensemble Knowledge	The implicit distribution over models that experts carry, built through years of practice, which they cannot fully articulate but which shapes their judgments. Single models presented publicly are communication artifacts standing in for this richer epistemic state.	This paper, III.1
Uncertainty Preservation Principle	The methodological principle that epistemic uncertainty should be maintained in its most structured form — distributions over models, not merely distributions over outcomes — and collapsed only when specific decisions require specific inputs. Equivalently: minimize informational compression until decision-forcing moments; request the least-compressed output your decision actually requires. Rests on epistemological foundations (uncertainty structure is information; compression destroys revisability) and connects to decision theory (Howard, 1966) and democratic accountability.	This paper, III.4
Underdetermination	The thesis that empirical evidence cannot uniquely determine which theory is correct, as multiple theories may be equally consistent with all possible observations.	Duhem (1954); Quine (1951)

Appendix H: Abstract Argumentation Semantics

Abstract Argumentation Frameworks (AAFs) provide formal foundations for the argumentation structures that AMTAIR extracts. While ArgDown captures structure and BayesDown adds probabilities, AAF semantics formalize which argument combinations are logically defensible.

Dung's Framework

Dung (1995) formalized argumentation as a directed graph where:

- Nodes = arguments
- Edges = attack relations (A attacks B means A provides reason against B)

Framework: $AF = (Args, Attacks)$ where $Args$ is set of arguments and $Attacks \subseteq Args \times Args$.

Semantics

Semantics define "acceptable" argument sets:

Conflict-free: Set S is conflict-free if no argument in S attacks another in S .

Admissible: Set S is admissible if: (1) S is conflict-free, and (2) S defends all its members (if X attacks $Y \in S$, then some $Z \in S$ attacks X).

Complete: Set S is complete extension if: (1) S is admissible, and (2) S contains all arguments it defends.

Grounded: Unique minimal complete extension (exists and is unique for finite AF).

Preferred: Maximal complete extensions (maximally admissible).

Application to AI Safety

AI safety debates map to AAFs. Example: Power-seeking AI argument structure:

- A1: "Advanced AI will pursue power"
- A2: "Alignment research will succeed"
- A3: "Power-seeking leads to catastrophe"
- A4: "Instrumental convergence makes power-seeking likely"

Relations:

- A2 attacks A1 (alignment prevents power-seeking)
- A1 attacks A2 (difficulty of alignment)
- A4 supports A1 (provides reason for power-seeking)

Note: Classical AAFs only model attack relations; support is handled through indirect attack (A4 attacks A2 because it undermines the alignment case).

Computing extensions reveals: Which combinations of beliefs are logically defensible given stated relationships?

Grounded extension: Arguments that must be accepted. Preferred extensions: Maximal consistent belief sets.

Integration with AMTAIR

AAF handles logical structure; Bayesian networks handle probabilistic quantification. Combined approach: Identify logically consistent belief sets, then evaluate probabilities within each consistent set.

Prokudin et al. (2024) showed AAF properties enable automated dispute resolution:

- Well-founded: Every AF has at least one extension
- Decidable: Extension membership computable
- Tractable: Linear time for certain structures (acyclic, bipartite)

Implementation via OWL DL ontologies enables automated dispute resolution in peer review — directly applicable to AI safety debates.

For extended treatment including Kuhnian paradigm analysis — where formal argumentation structures may support the transition from pre-paradigmatic fragmentation to constructive scientific consensus (Kuhn, 1962) — and cultural evolution frameworks, see Meyer (2025b) (Appendices B-E).

Appendix I: Critical Analysis Extended

This appendix provides detailed technical discussion of limitations and anti-gaming mechanisms summarized in Section V.

Epistemic Culture Clashes

AMTAIR's Bayesian formalism aligns more naturally with some research traditions than others. Formal verification approaches prioritize deductive certainty; empirical ML research prioritizes observed performance; policy analysis prioritizes actionability. These are genuinely different epistemologies, and AMTAIR's probabilistic framework creates systematic representation gaps for traditions that operate differently. The parallel to economics — where rationalist and empiricist traditions face persistent tension — is direct. This does not invalidate the approach, but honest implementation must acknowledge where its formalism fits poorly.

E/P/I as Emphases: A Closer Look

A tension worth examining: the paper describes arguments as having "explanatory emphasis" or "predictive emphasis," but Veit's E/P/I distinction concerns *epistemic functions* (what models do in context), not stable properties of arguments. The same model can serve explanatory purposes in one context and predictive purposes in another. The case studies in Section IV.5 respect this — the Carlsmith network supports all three query modes regardless of its emphasis — but readers should resist treating E/P/I labels as categories attached to arguments rather than modes of inquiry applied to them.

Systematic Catalog of Epistemic Losses

Table 4: Epistemic content preserved and lost by BayesDown conversion, organized by argument emphasis. Note: each extracted network supports all E/P/I query modes regardless of emphasis.

Argument Emphasis	What BayesDown Preserves	What BayesDown Loses
E-emphasis (e.g., Carlsmith)	Probability estimates for conjunctive conditions; causal structure as DAG	Mechanistic depth explaining <i>why</i> conditions hold; philosophical argumentation quality; rhetorical force; counterfactual reasoning beyond explicit conditional probabilities
P-emphasis (e.g., Scaling Laws)	Quantitative forecasts; uncertainty ranges; conditional dependencies	Model provenance and training assumptions; informal domain expertise; contextual caveats; sensitivity to distributional shift
I-emphasis (e.g., Dafoe)	Policy recommendations as probability-weighted outcomes; causal intervention effects	Political feasibility assessments; implementation pathway knowledge; institutional constraints; normative framings

Anti-Gaming Mechanisms

AMTAIR incorporates several friction mechanisms designed to resist strategic manipulation:

Adversarial Red Team Simulation. Before any model enters the ensemble with significant weight, it undergoes adversarial review by designated skeptics. The red team attempts to: (1) construct alternative interpretations yielding dramatically different estimates; (2) identify unstated assumptions that would change conclusions; (3) find internal inconsistencies; (4) propose modifications that would flip conclusions.

Sensitivity-to-Prior Score. Each model receives a "sensitivity score" measuring how much its influence depends on prior specification:

$$S_k = \max_{\pi \in \mathcal{P}} \left| \frac{\partial R(\pi)}{\partial P(M_k)} \right|$$

High-sensitivity models face mandatory documentation explaining why their prior assignment is justified.

Validation Time Proportional to Weight. Models approaching significant influence (>5% of ensemble weight) receive proportionally more expert review time.

Structural Complexity Penalties. Arguments that are suspiciously well-formatted for extraction receive penalty flags — unusual structural simplicity triggers additional review.

Tacit Knowledge: A Nuanced View

A counterargument deserves consideration: perhaps tacit knowledge is less inaccessible than the main text suggests. Large language models are trained on vast corpora where implicit judgments leak into linguistic patterns. When AMTAIR extracts from arguments written by experts, it may capture shadows of tacit knowledge that diffused into language.

This is genuine but limited. Three categories remain genuinely inaccessible: (1) *classified or proprietary information*; (2) *embodied knowledge* from hands-on system interaction; and (3) *social knowledge* circulating through private channels.

When Robustness Analysis Yields No Robust Policies: Extended Discussion

What if policy preference mapping reveals *no* interventions that perform well across models? If different models recommend incompatible policies — scaling-focused models prioritizing compute thresholds, alignment-focused models prioritizing interpretability requirements, coordination-focused models prioritizing international agreements — has AMTAIR not failed to solve the coordination problem it was designed to address?

Absence of robust policies is *informative*, not a failure. Without systematic analysis, policymakers might adopt whichever framework speaks loudest, unaware that other credible models recommend incompatible approaches. The false confidence of acting on one model while ignoring others creates fragility — the policy may fail precisely because it optimized for assumptions that turn out wrong.

When AMTAIR reveals no policy-level convergence, it has performed exactly as designed: it has surfaced a genuine disagreement that demands attention rather than papering over it with premature consensus. The appropriate response differs by case:

- **Disagreement about facts:** If models diverge because they make different empirical predictions (e.g., capability timelines), the disagreement identifies crucial uncertainties where additional research or forecasting would have high value of information.
- **Disagreement about values:** If models diverge because they weight outcomes differently (e.g., economic disruption vs. existential risk), the disagreement is fundamentally political, requiring democratic deliberation rather than technical resolution.
- **Disagreement about mechanisms:** If models diverge because they posit different causal pathways, the disagreement suggests targeted empirical investigation or scenario planning for multiple contingencies.

In each case, *knowing* that models disagree — and *understanding where* they disagree — enables more sophisticated governance than false confidence would permit. A policymaker who knows "these

three interventions are robust, but these two are contested" can act on the robust subset while investing in resolving the contested questions. This is not paralysis; it is calibrated action under acknowledged uncertainty.

The worst outcome is not discovering that models disagree. The worst outcome is acting as if they agree when they do not — and discovering the disagreement only when policies fail.

Appendix J: Technology-Enabled Philosophy

This appendix expands on the philosophical significance of AMTAIR's emergence at a specific technological moment.

Instruments and Inquiry

Scientific instruments enable new forms of inquiry. Telescopes did not merely improve astronomical observation; they made possible questions that could not previously be asked. Microscopes transformed biology from taxonomy to mechanism. Computers enabled mathematical explorations (chaos, complexity, simulation) inaccessible to pencil-and-paper methods. Each instrument opens a new space of possible investigations.

LLMs represent a similar inflection for certain philosophical practices. The conceptual infrastructure developed in this paper — level-separation, uncertainty preservation, multi-functional queryability — could be *articulated* before frontier LLMs, but could not be *practiced* at scale. Earlier NLP lacked the semantic understanding to extract causal structure from complex argumentative prose; earlier argument mining produced claims without probabilistic enrichment. AMTAIR exists because current models can parse nuanced reasoning, identify implicit conditional relationships, and generate coherent probability estimates from qualitative hedges. This is technology-enabled philosophy: the instrument makes the practice possible.

Reflexivity and Shared Failure Modes

A reflexive dimension adds complexity: AMTAIR uses AI systems to formalize reasoning *about AI systems*. The same technological trajectory that creates the governance challenge — increasingly capable AI with uncertain alignment properties — provides tools to address it. This is fortunate (new capabilities) and concerning (shared failure modes). LLMs analyzing arguments about LLM risks may have systematic blind spots — training biases about AI capabilities, difficulty recognizing failure modes they themselves exhibit.

Theory-Ladenness of Extraction

More fundamentally, LLMs are not neutral extraction instruments. Their training data, architectures, and RLHF embed assumptions about valid reasoning, appropriate probability assignments, and causal versus correlational relationships. AMTAIR inherits these assumptions. A pipeline built on different LLMs might produce systematically different formalizations. The extraction is theory-laden in the sense Kuhn (1962) identified for observation — shaped by prior frameworks, never purely given.

None of this invalidates the approach. Telescopes also introduce distortions; statistical software embeds modeling assumptions; every instrument shapes what it reveals. The point is that AMTAIR's emergence at this technological moment is not incidental to its epistemology. Understanding its capabilities requires understanding the instruments that make it possible.

Appendix K: Cross-Paper Dependencies

This paper (Meyer, 2025d) is one of three companion papers exploring different dimensions of AMTAIR. Each can be read independently; this appendix traces the substantive connections.

Relation to the evolutionary epistemology paper (Meyer, 2025b). That paper diagnoses the problem this paper's epistemological framework addresses. EvoRat demonstrates through cultural evolution theory that AI governance discourse systematically selects for memetically fit arguments over epistemically accurate ones, creating information cascades and coordination failures. This paper asks: given that diagnosis, what epistemological stance should AMTAIR adopt? The answer --- model pluralism over premature selection --- is grounded in the philosophy of economics tradition (Cartwright, Rodrik, Veit) but motivated by EvoRat's finding that premature convergence is the specific failure mode to guard against. EvoRat's Appendix H.6 traces how acceptance of this paper's level-separation framework would strengthen EvoRat's decomposition claim (P9) and values objection defense (D4).

Relation to the organizational epistemology paper (Meyer, 2025c). That paper operationalizes the statistical infrastructure this paper's epistemological arguments justify. This paper argues that maintaining multiple competing models is epistemically superior (Section II); EpiOrg asks how to do so in practice when integrating prediction market data that could contaminate model independence. The Bayesian Model Averaging framework developed in Appendix B of this paper provides the mathematical foundation for EpiOrg's quality-weighted updating. EpiOrg's contamination taxonomy (Section III) in turn reveals a practical constraint on pluralism: maintaining genuine model diversity requires active contamination prevention, not just philosophical commitment.

What this paper contributes to the others. This paper provides the epistemological foundations the other two presuppose. EvoRat's prescription --- that AMTAIR should enable "epistemic clarity for normative negotiation" --- rests on the claim that empirical disagreements can be separated from normative ones. This paper grounds that claim in Cartwright's nomological machines and Veit's model pluralism, showing how formal models can coexist productively rather than collapsing into a single paradigm. The ArgDown/BayesDown specification (Appendix A) and the complete Carlsmith extraction serve as shared technical reference across all three papers, demonstrating that AMTAIR's formalization is feasible and epistemologically defensible.

Acknowledgements

I thank Prof. Walter Veit for supervision of this work and for developing the model pluralism framework that this paper extends to governance practice. I also acknowledge the broader AI safety research community whose arguments provided the empirical material for AMTAIR's development and validation. Special thanks to friends and family who provided invaluable feedback on earlier drafts, helping clarify arguments that seemed clear only to me and identifying gaps that expertise had rendered invisible. Any errors in the extension or interpretation of philosophical concepts remain my own.



UNIVERSITÄT
BAYREUTH

– P&E Master's Programme –
Philosophy & Economics
University of Bayreuth

Affidavit

Declaration of Academic Honesty

Hereby, I attest that I have composed and written the presented paper independently on my own, without the use of other than the stated aids and without any other resources than the ones indicated. All thoughts taken directly or indirectly from external sources are properly denoted as such.

This paper has neither been previously submitted in the same or a similar form to another authority nor has it been published yet.

BAYREUTH, December 31, 2025

VALENTIN MEYER